

Medical Applications

The medical community actively seeks new technologies to enable them to provide better medical care. Tools such as the stethoscope, x-ray machine, microscope, and many more have enabled doctors to provide better patient care, medical research, and enhance and prolong life. While some tools have been specifically developed for medical uses, others that were not developed specifically for medicine are applied with positive results. Virtual reality applications for medical use are being developed and tested for a wide variety of tasks. While most applications are still in the development and testing phase, others have already been deployed for production use.

It is interesting to note that a significant amount of funding for research in medical applications of virtual reality has been provided by the U.S. Navy. The Navy sees promise that virtual reality can be a useful tool in support of their work in the medical care of our troops while they defend our country. In addition to the enhancement of more conventional medical care, they are also interested in applying virtual reality techniques to a more specific area of interest called "telemedicine." Telemedicine is a combination of telepresence, augmented reality, and virtual reality focused on allowing physicians to diagnose and treat patients at a distance. Telemedicine offers the promise of bringing high-quality doctors into a war zone that may be thousands of miles away.

Although many compelling demonstration applications have been created, relatively few have made it from the laboratory into daily production use. There are a number of reasons that account for this. Among them are the rigorous testing and government approval that new technologies are required to go through prior to adoption. Also, the medical community is largely very conservative, so they are often slow to migrate from the tried and true to the new and different. Many physicians are skeptical by nature, and have little time to devote to learning new techniques unless they are convinced the effort will pay off.

In order to have a high likelihood of adoption, a medical application of virtual reality must provide a benefit that outweighs the costs associated with adoption. One is that the system must be easy to learn and relate to techniques and procedures that the physician is accustomed to, while providing new capabilities. Later in this chapter, we cover the University of Virginia's *Neurosurgical Visualization* application which specifically addresses this concern.

Applications must provide accurate results, sufficient resolution, and any sensory output appropriate to the task. Because touch is a critical element in many medical procedures, haptic feedback is often important. Likewise, the use of olfaction in virtual reality is likely to occur in medical applications in the future.

5.1 AREAS OF APPLICATION

Virtual reality is being used in a variety of ways in the medical field ranging from drug design, to empathy training (see *Detour* in Chapter 8) to Phobia treatment (see *Fear of Flying* later in this chapter) There are four primary areas of medicine that are covered here. In each of these areas, there is a broad division of whether applications are primarily focused on supporting medical research or focused on clinical use. These application areas include:

- Training
- Scientific visualization
- Preoperative planning
- During-operation assistance

New application areas continue to emerge. Virtual reality is being applied in dentistry, otolaryngology (ear, nose, and throat), balance dysfunction, training for patient interviews, endoscopy, and other fields. The medical community is sufficiently interested in virtual reality that there is a conference specifically dedicated to VR in medicine. The "Medicine Meets Virtual Reality" (MMVR) conference provides a venue for interested participants to exchange ideas, view the latest products, and publish important findings. The MMVR conference has been active for over 15 years.

CASE STUDY 5.1: BDI SURGICAL SIMULATOR

Executive Summary

When medical students learn the complicated process of surgery, they use all of their senses. In contrast, many virtual reality applications rely primarily on visual and auditory information. Surgery, however, relies very heavily on the sense of touch. A surgeon reacts not only to what they see, but also what they *feel*.

In this focus, Boston Dynamics Inc. tackles the issue of the sense of touch in their turnkey surgical simulator system. Their system uses actual surgical instruments in tandem with a visual display housed in a kiosk that emulates the actual surgical environment. The instruments are actively manipulated by the VR system to provide realistic haptic feedback to the student. The system also evaluates the



FIGURE 5-1

A surgical student wears shuttering stereo glasses to look into a virtual body cavity while practicing the surgical procedure of anastomosis.

Image courtesy Boston Dynamics, ©1997–1998

student and provides a report of their progress in learning new skills.

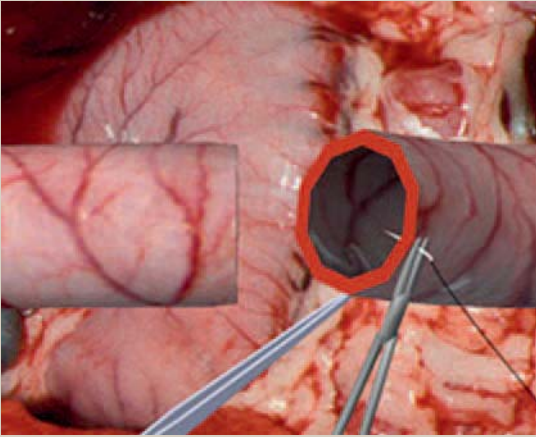
Introduction

The BDI *Surgical Simulator* is a medical training tool that uses force feedback to give life-like responses to a specific surgical procedure. Boston Dynamics Inc. (BDI) has been involved in creating realistic (physically based) motion dynamics simulations and haptics applications. The company slogan—“Focusing on the Science and Art of How Things Move” reflects their emphasis in applying real-time motion simulation to tasks that can be significantly enhanced by such technology. BDI has also used this technology to develop applications related to aircraft maintenance training and design and human factors analysis. Here, we will focus on their surgical simulator for training open surgery and anastomosis tasks.

The *Surgical Simulator* product is an integrated set of hardware and software utilizing VR techniques to provide an

experience which represents reality within a set of commonly used surgical procedures. Of particular interest is the usage of haptic feedback as a primary form of display. In fact, the haptic feedback is based on user-position data, whereas the visual display is not, thus making the haptic display a significant factor by which the participant is physically immersed.

The specific goal of the *Surgical Simulator* is to re-create the experience of doing “end-to-end anastomosis”—the task of suturing tube-like organs together. This task was chosen partially because anastomosis is found in many surgical procedures involving various organ systems, such as blood vessels, esophagus, colon, and bile ducts. BDI designed the simulator to accommodate these various organs by providing an adjustable set of parameters, such as tube diameter, tissue stiffness, surface texture, surface friction, needle piercing forces, and so forth. Models of the different tissues and tube organs are created to both look and feel like the real thing.

**FIGURE 5-2**

Body organs are rendered as realistically as possible within the constraints of real-time interaction. In this image a student uses a needle holder and forceps to practice end-to-end anastomosis.

Image courtesy Boston Dynamics, ©1997–1998

Application Description

In the *BDI Surgical Simulator*, participants hold actual medical instruments to practice surgical tasks in a virtual world. This allows the medical student to practice without the use of hospital space, human patients, animals, or constant supervision by a surgeon.

In the simulator, medical students practice the various aspects of end-to-end anastomosis by using forceps, a needle, and a needle driver to touch, grasp, and suture two compliant tubular organs. Actual medical instruments are used as the interface to the virtual world. A student grasps the real instrument, and in the computer graphic view can see a realistic avatar representation.

The surgical instruments are connected to force-feedback devices to provide haptic feedback to the student. Haptic feedback is an important aspect of performing surgical procedures. Simulators like this can allow students to encounter unexpected conditions and rare medical disorders, which they might not have the opportunity to encounter in standard training methodology. The simulator allows the student to practice and be prepared for either routine

**FIGURE 5-3**

Actual medical instruments such as forceps and needle holders are attached to force-feedback devices. The end result is that the student can feel their interactions with virtual organs. They can sense the elasticity and stiffness of tissue and can sense when a needle penetrates an organ.

Image courtesy Boston Dynamics, ©1997–1998

or unusual procedures. To use the surgical trainer, the student grasps the forceps in one hand, and the needle driver in the other. Using the actual forceps, the student grasps a virtual tube in one hand, and punctures it with the virtual needle held by the needle driver in the other hand. The student then sutures the tubes together by puncturing the other tube with the threaded needle and then pulls the suture tight. The actual knot tying is performed automatically by the simulator, as there are other effective ways to teach this skill.

In conjunction with the *Surgical Simulator*, BDI has also developed a simpler system created to measure task performance on specific actions such as time to completion, needle placement accuracy, tissue surface damage, tangential tearing force, and others.

Representation of the Virtual World

The content of the virtual world is the small work volume of normal open cavity surgery, including the organs involved in the procedure and the instruments with which the user operates.

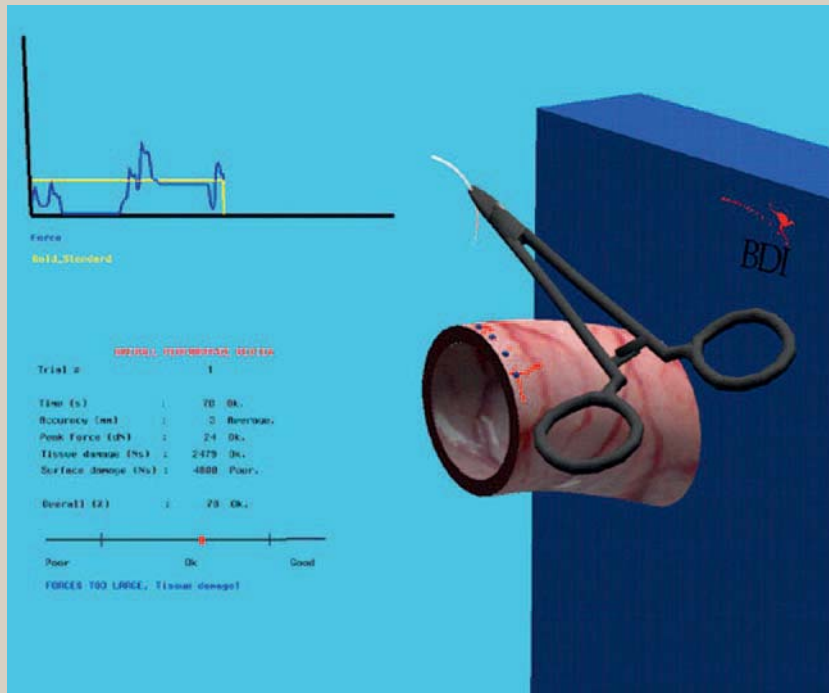


FIGURE 5-4

A simpler skills assessment simulator allows students to practice subtasks of anastomosis (e.g., pushing the needle through at well spaced intervals without damaging too much tissue), plus provides feedback on how well they did.

Image courtesy Boston Dynamics, ©1997–1998

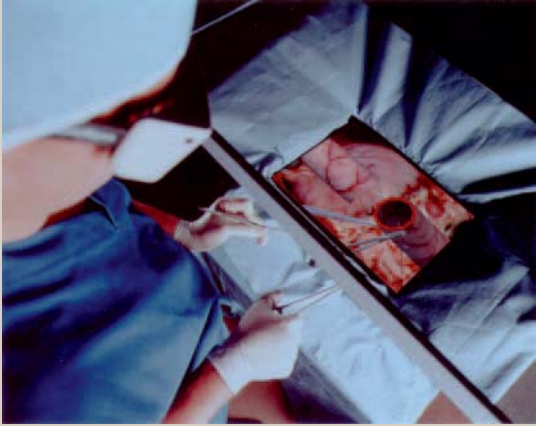
The world is visually and haptically represented as realistically as feasible for the task. The graphics and haptics are driven by a fairly realistic dynamic model in real time. However, constraints on software development limited how many features could be incorporated as part of the initial product. For example, blood and other fluids that normally obstruct a surgical procedure are not rendered, resulting in a much cleaner opening than a surgeon would normally encounter.

While students are performing the task and feeling the results via the instruments, they are also able to see what they are doing on a visual display arranged to mimic looking into the patient's open cavity. In the visual display, they can see the deformation of the tubes and organs. In real life, there can be obscuring factors such as blood and

other tissues, as well as a constricted maneuvering space. Some of these factors are being developed for integration with training scenarios.

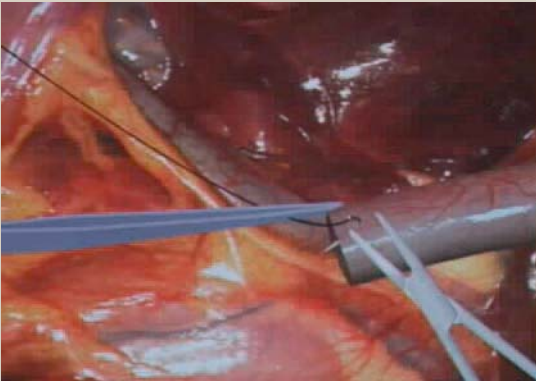
The world is visually rendered with a realistic visual representation of the tissue surfaces using texture maps. Two open-ended tubular structures can be seen at either end of the operating area, with a view into the body cavity below. Also included in the computer-generated scene are avatars of the medical instruments. The instrument avatars move in correspondence with the movements of real instrument props that the student manipulates.

Ordinarily, the student will see their own hand holding one end of the instrument, and the other end of the instrument represented via computer. If they put their hand far

**FIGURE 5-5**

Avatars of the forceps and needle holder are evident as the student practices suturing.

Image courtesy Boston Dynamics, ©1997–1998

**FIGURE 5-6**

Haptic feedback devices are hidden from view by a mirror in which the surgical student sees the virtual body cavity and organs.

Image courtesy Boston Dynamics, ©1997–1998

enough into the scene, then the entire instrument could be viewed, but as though floating in space. When the application designers added hand avatars in the scene, experienced surgeons found the avatars to be distracting, so they were removed from the final product.

There is no audio representation in the surgical simulator other than the natural sounds made by the physical instruments held by the student.

The haptic representation is perhaps the most notable aspect of this application. Force-feedback systems are connected to the two medical instruments—a needle holder and a forceps. The trainee uses both instruments simultaneously to manipulate and feel the environment. Using the instruments, the student can grab the tissue, drive the needle through the tissue, and bring the thread together to generate a knot. As they pull the tissue, they can feel its resistance to stretching, and as they puncture the tissue with the needle, they can feel the resilience, and then the pop as the needle glides through.

To create realistic feedback on the medical instruments, a real-time dynamics model was incorporated using physical models of tissue stretching and tearing. This model can be adjusted with parameters to simulate tubular structures of different thicknesses and tissue type.

For more complicated surgical training tasks, one might imagine the use of smell to be a clue for detecting substances that might otherwise be undetectable, such as necrotic tissue. Much like audio often provides a great enhancement for mental immersion in a space or task, in a surgical environment, smell may be just as powerful, if not more so. There is no need for vestibular feedback, as the task itself is performed with the surgeon standing over the patient, and not moving into very many orientations.

Interaction with the Virtual World

The BDI *Surgical Simulator* can clearly be classified as an experiential *training* application. The form that the interaction takes is that of mimicking the real-world task where the surgeon stands next to the patient lying on a table and manipulates surgical implements.

The procedure followed by the students using the *Surgical Simulator* is a subset of actual surgical procedures. As much as is feasible, the students are made to feel they are working under normal operating conditions.

There is no travel necessary in the simulator beyond the physical locomotion of the hands. It is entirely through the

use of the hands manipulating the prop instruments that the user interacts with the world.

There is no true interaction with other users or with autonomous agents in the virtual world. The only other being that is represented is assumed to be anesthetized.

The simulator is an attempt to represent the real world and is based on a real-time dynamics model to produce responses that reflect real-world laws of nature. This includes some collision detection necessary for determining when the instruments make contact with the organs as well as when the organs make contact with one another.

The deformable tube simulator is optimized for real-time performance. The simulation supports tube bending and stretching, as well as local surface deformation such as dimpling and puckering. The simulation allows parameters of the tubes such as length, diameter, thickness, mass, surface friction, compliance, and resistance to puncture to be varied to allow the system to represent different anatomical elements. The system parameters were adjusted by tweaking them until the results felt “correct” to experienced surgeons. This technique led to a more realistic “feel” than trying to take measurements from actual tissue.

Physical characteristics that are not supported include tissue folding, contact against puckered or dimpled regions of tube, or layers of tissue.

In order to approach the desired rate of 1000Hz for interactions with rigid objects (soft tissue and other compliant objects are less demanding), the number of possible contacts between different objects was limited. One contact, for example, that was not computed was the instruments contacting each other.

Venue

The surgical simulator is based on a kiosk-style stand that closely resembles the station at which a surgeon would perform an operation. The typical location of this kiosk would be a medical school or hospital.

After a session with the full simulator, or the skills assessment package, a “Surgical Report Card” is delivered to the student.

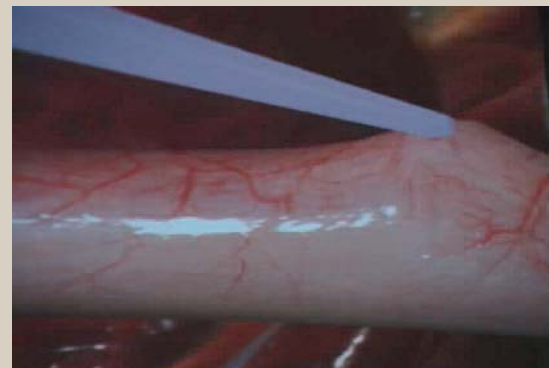
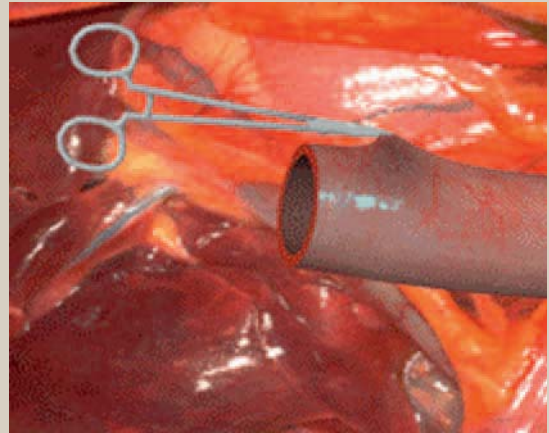


FIGURE 5-7

Physical characteristics of tissue, such as elasticity are modeled and rendered both visually and haptically. Here the student can feel the tissue stretch under their grasp.

Image courtesy Boston Dynamics, ©1997–1998

VR System

To view the graphical scene, the student stands at the kiosk looking downward toward a mirror that reflects the visual display from above. The reason for the mirror is to mask the mechanics of the haptic devices from view. The student then holds onto the instruments beneath the mirror. Cotton sheets disguise the monitor, in much the same manner that they hide the rest of the patient from view during an operation. As available during actual operations,



FIGURE 5-8

The BDI Surgical Simulator is a turnkey system that is housed in a kiosk. The kiosk provides hand rests and resembles an actual surgical station.

Image courtesy Boston Dynamics, ©1997–1998

a hand rest is provided on the kiosk to allow the surgeon to stabilize their hands.

The medical tools provide a prop-style interface to the virtual world. The tool “props” are instrumented to sense the amount of closure or clamping pressure is being applied by the user. To sense the position and provide force feedback to the user via the props, they are attached to haptic I/O devices.

Specifically, two SensAble Technologies’ PHANTOMs™ are used, one for each instrument.

The student wears untracked shuttering glasses to allow them to see the stereoscopic visual representation. Although the head is not tracked, the developers felt the benefits (motion parallax) would not outweigh the costs (latency and instability in viewing). The application does qualify as a VR application, because the hands are tracked (via the instruments), and the virtual world responds directly to those actions.

The hardware components consist of an SGI Maximum Impact Indigo II workstation to compute the graphics. A PC runs the simulation and controls the haptic feedback devices. A PHANTOM device for each hand provides the haptic feedback.

The graphical software for the system is written using the *OpenGL* library. The scene consists of about 1200 polygons and is rendered and displayed at about 20 Hz. Other software consists of modules to perform the deformable tube simulation, fast contact detection, contact force calculation, the kinematics for the force feedback devices, and the force-feedback device drivers.

Application Implementation

Surgical Simulator required over 3 years to develop, representing over 12 man-years of effort. Funding for developing this product was provided by Dr. Rick Satava of DARPA, and Dr. Tom Krummel of the Pennsylvania State Hershey Medical Center. The application was created as an integration of a dynamic simulation of human tissue properties, stereoscopic display, and haptic feedback. An interesting aspect of this application is that actual surgical tools were used rather than more generic props.

Conclusion

Although the BDI *Surgical Simulator* does not fully replicate all the details involved in an actual surgical procedure, the system designers feel that it is a significant step toward that goal. The system has been tested with a variety of medical students, residents, and surgical instructors, including a few well-known surgeons. The response has been quite positive. Many of those who have tried the

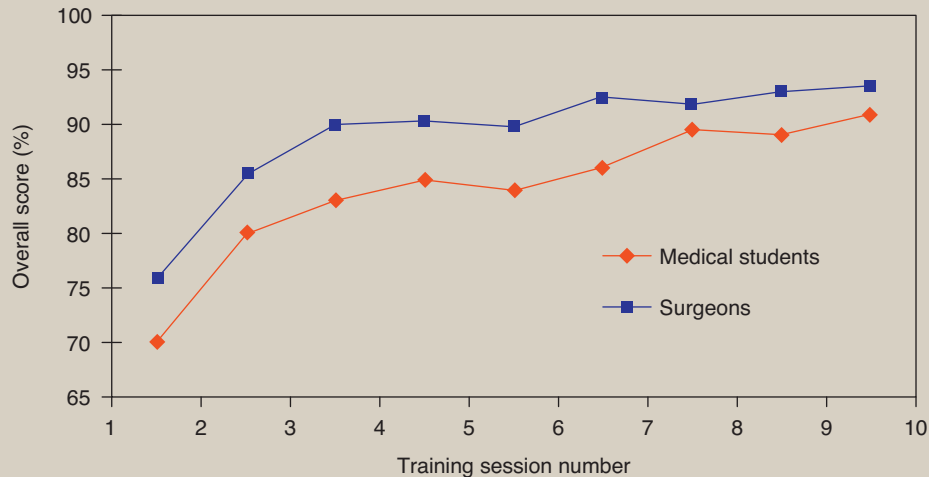


FIGURE 5-9

This graph shows that while experienced surgeons have a higher overall skill level than student surgeons, both groups improved their skills by using the VR Surgical Simulator.

Image courtesy Boston Dynamics, ©1997–1998

simulator are of the opinion that it would be a valuable tool in surgical training.

The results of these users (12 students and 9 surgeons) have been compiled into an informal study (overall results shown in Figure 5.9). Two observations from the overall results are that there is a marked difference between the skills of the surgeons and those of the students, and each group showed improvement over the eight trials.

The use of virtual reality in surgical training enables surgeons to be better prepared, and more skilled, as they prepare for real-world procedures. The *Surgical Simulator* may also prove useful to veteran surgeons by enabling them to practice for a rare or unusual cases prior to an actual procedure.

In closing, Marc Raibert, president of BDI, says: “Surgical simulators revolutionize how surgery is taught and how surgical skills are measured. The BDI Anastomosis Simulator is a significant step in that direction.”



FIGURE 5-10

BDI has developed another simulator that is used for training students to do knee arthroscopy procedures.

Image courtesy Boston Dynamics, ©1997–1998

CASE STUDY 5.2: CELIAC PLEXUS BLOCK SIMULATOR

A **celiac plexus** block is a medical procedure for injecting a drug in a particular region within the torso of a patient. This procedure is carried out by inserting a needle into the back of a patient who is lying on their abdomen, and injecting a drug in the appropriate location. Finding the precise location of the needle inside the body is a difficult skill to learn. The physician uses a combination of their knowledge of anatomy with periodic x-ray views displayed on a fluoroscope. The physician uses their sense of touch when directing the needle through the skin, soft tissue, and other anatomic objects.

A problem that can occur during the procedure is injecting the drug in the wrong location, which can have catastrophic effects (if injected in a major blood vessel, for example) or, more likely, cause the procedure to be ineffective. Another potential problem is damaging good nerves as a result of repeated repositioning of the needle.

Researchers at the Center for Human Simulation at the University of Colorado Health Sciences Center have developed a simulator for training physicians in this procedure. The primary benefit of this VR application is the accelerated learning possible with the simulated procedure.

Application Description

The medical procedure supported by this VR application is normally done by having the patient lie on their stomach. The physician chooses a location on the back as the needle entry point best suited for delivering the drug to the target area. The patient is typically given a local anesthetic at the point of entry of the needle, but is normally awake during the procedure. As the physician inserts the needle, they can feel the different tissues as it moves through the body. When they feel that the needle tip is at the appropriate location, or need help determining its location, they can request a fluoroscopic update.

Because the actual procedure relies so heavily on the haptic feedback provided by the needle encountering the various parts of the body, the simulator provides an excellent platform in which to utilize haptic-feedback devices.

The simulator closely replicates the environment the physician would encounter in their actual practice. The elements



FIGURE 5-11

A user grasps the end of an actual needle (controlled by a force-feedback device inside the back) to practice the celiac plexus block procedure and can view the results on the nearby simulated fluoroscope display.

Photograph courtesy of Karl Reinig

that are readily visible to the physician include a life-size physical model of a human back, an actual needle identical to ones used in the procedure, and a computer monitor that mimics the display of the fluoroscope.

The *Celiac Plexus Block Simulator* attempts to mimic as much of the actual procedure as necessary to convey a realistic sensation of performing the operation on a human

patient. During the simulation session, the physician can use the integrated “virtual fluoroscope” to display x-ray views of the patient, including the location of the needle on the computer monitor. Although the monitor is smaller than an actual fluoroscope, it is closer to the physician, with the net result being a realistic simulation. The system is capable of delivering continuously updated views on the display. However, in the real-life scenario the physician is only able to get updated views periodically, so this constraint is maintained in the simulator. During the actual procedure, each time the physician requests an updated view, everyone but the patient must leave the room while the image is being taken. This can be replicated with the simulator.

In this simulator, the PHANToM™ from SensAble Technologies provides the force feedback. The PHANToM itself was hidden from view, housed within the shell of the virtual patient’s back. This simulator is a good example of how technology can be hidden from the user, thereby decreasing the likelihood of the illusion being broken by its physical presence. When the doctor inserts the needle into the virtual patient, the PHANToM provides forces on the needle that cause the physician to feel the needle penetrate the skin, soft tissue, be obstructed if it encounters bone, etc. Additionally, the physician can feel the pulsing of blood flow when the needle is positioned near or in a major blood vessel.

The prototype system uses the data from the Visible Human Dataset.¹

Additional patients could be modeled from MRI and CT data. However, it would be a tedious process since every volume element of the data (voxel) needs to be segmented and classified in order for the system to determine the appropriate force feedback to provide. If each voxel in the area the needle is penetrating is not classified, the trainee cannot make (and recognize)

all the mistakes that should be allowed in a useful simulator.

While the interactively controlled needle provides fairly realistic haptic feedback based on user input, the back—which is modeled out of fiberglass—is inflexible, and thus not quite as haptically realistic. The inflexible back is also counter to the notion of a dynamic body in which the physician is positioning the needle. Internally, the application does simulate some dynamic tissue movements within the body. For example, the aorta changes in correspondence to a beating heart. They plan to implement more internal movements in future versions, such as the movement of the diaphragm to replicate the breathing of the virtual patient.

In addition to the haptic and visual sensory displays, the *Celiac Plexus Block Simulator* also provides aural feedback cues. In particular, if the user attempts to push the needle into a bone with forces that would cause pain in a patient, the system triggers the playback of a digital recording of the sound of a cry of pain. The expert advisors to the application developers provided input as to how hard a push into the bone would need to be to cause significant pain in a patient.

The fiberglass shell, representing the patient’s back, was created by using the visible human dataset to drive a Dremel™ grinding tool attached to the PHANToM. They then cut the back from an alpha cradle block (the same expanding foam used to restrict the motions of patients in CT or MRI scanners). The block was then used to create a plaster mold from which the fiberglass model was made. The current model is positioned upright as though the patient were seated. The PHANToM can be reoriented to lay the model down as though the patient were on their abdomen, more closely replicating the typical real-life scenario.

The model of the back was created from the Visible Human Dataset, also created at the University of Colorado’s Center for Human Simulation (UC-CHS). The same dataset was used to model the internal organs of the simulated patient. Some modifications were made to the dataset to correct artifacts that were introduced as a result of the procedure used to acquire the dataset.

¹The Visible Human Project produced large, high-resolution datasets of a male and a female body using multiple imaging techniques [Spitzer et al. 1996].

The project took about four months to develop with the creators working approximately half time on it. Karl Reinig (a post-doctoral fellow at the UC-CHS) came up with the idea after encountering a haptic feedback device. Reinig was responsible for the basic design and haptic algorithms. Kenneth Niedjadlik, MD, Christopher Loeffler, MD, and Jose Angle, MD, were the anesthesiologists who lent their expertise to the program by teaching the developers about celiac plexus blocks, and conveying how the needle insertion should feel. Helen Pelster segmented and classified the visible human data to make it work for this project. Chuck Rush developed the PHANToM-NT driver, the application GUI, and helped Reinig develop the overall hardware and software system. Jim Heath helped Reinig develop the method for creating the fiberglass shell. Extensive use was made of data from the Visible Human Project led by Victor Spitzer, PhD, and David Whitlock, MD.

VR System

The primary hardware used by the *Celiac Plexus Block Simulator* is the PHANToM™ haptic device. The PHANToM—mounted inside the physical shell of a molded or carved back—is connected to the end of an actual needle that would be used in this type of operation. The PHANToM provides resistance to the motion of the needle. The simulation (including the PHANToM driver and fluoroscope graphics) was run on an IBM-compatible, Pentium Pro™-based personal computer. Sounds, such as the “yelp” of pain, were generated by loading WAVE sound sample files into memory and playing the sample when a specific event occurs.

A footswitch provides an emergency stop control for the PHANToM in the event of a software error, or other anomaly that may cause the PHANToM to go out of control. The consequences of a runaway condition in a haptic display can be more severe than in other displays.

The Center for Human Simulation also involved in several other VR applications. The *Dental Probe Simulator* uses the PHANToM in conjunction with a three-dimensional graphical display to allow a student to practice and be tested on their ability to diagnose tooth decay. They use the PHANToM to simulate a dental explorer tool. Rather than using an actual dental explorer, they use the generic

PHANToM probe extension. The tool can be constrained to only allow the movements that would be allowed in the real world in the patient’s mouth cavity.

The data is created from CT scans, and caries are artificially added to the tooth model. The participants feel the soft area, the movement of the jaw when pressure is applied, and the resistance to the removal of the explorer tool.

Another project is a *Scalpel Simulator* that allows the student to practice cutting through different types of materials, including fat (easy), tendons (a little harder), bone (very difficult), etc. This application uses the PHANToM for the force feedback, and an SGI for the 3D, texture-mapped graphical display.

Later, the UC-CHS VR team developed what they call the *Haptic Workbench* to create medical-related VR applications requiring force feedback. Mirrors are used to collocate a 3D visual image with the haptic devices. The haptic devices can be used to mimic a variety of operating tools such as a scalpel, a palpator, a needle, or a scope, all of which can be used to allow the user to interact with any part of the anatomy rendered on the screen.

One demonstration application they developed with the *Haptic Workbench* used two haptic devices to practice a distal targeting of a femoral nail procedure. In this procedure, a drill is hammered into a hole in a titanium shaft planted in the femur. If the drill is not lined up properly, then no amount of hammering will advance the drill into the hole. The two PHANToMs allow the user to line up the drill bit with the virtual patient’s leg. The *Haptic Workbench* display provides both a 3D view and a fluoroscope display of the leg and tools. The benefits of simulating such a procedure extend not only to the patient, but also to the doctors because they are not exposed to harmful x-rays while practicing the operation.

Conclusion

Many of the doctors who have had the opportunity to try it have found it to be quite accurate. While the lab has not continued to work on the *Celiac Plexus Block Simulator*, they have remained active in pursuing other possible uses of virtual reality for simulation and training of medical procedures.



FIGURE 5-12

A physician practices a femoral nail procedure on the Haptic Workbench.

Photograph courtesy Karl Reinig

The *Celiac Plexus Block Simulator* is a pioneering example of multimodal representation in a medical VR application. The sense of touch conveys a great deal of information to a physician when performing this procedure. The *Celiac Plexus Block Simulator* used several forms of haptic feedback to provide a meaningful experience to the physician practicing with the VR application. Real world medical instruments were used as props to allow the physician to feel the actual mechanics of those medical tools.

A PHANTOM device provided force feedback based on the type of tissue that was being penetrated. The shell of the body of the patient also provided haptic feedback, though not with a great deal of haptic verisimilitude to a patient's actual back. Sound was also used to emulate sounds that a patient might make during the procedure. In fact, the haptic components of this application were the elements that set this application apart from other similar applications at the time. The visual feedback was very similar to that conveyed during an actual procedure. An interesting aspect of this application was that while the hands of the physician were tracked via the instruments, there was no tracking of the head in this application. Hence, this application provides immersion via haptics rather than from the visual display. In summary, the combination of visual, haptic, and auditory feedback combined enabled a new type of simulator to be created that allowed physicians to practice a procedure they currently couldn't practice without an actual patient.

CASE STUDY 5.3: AUTISM THERAPY

Government regulations guarantee that all children in the United States receive an education. This includes children with special disabilities and handicaps. Unfortunately, current educational practices do not always meet the requirements of all children with special needs. Computer engineer Dorothy Strickland led a joint project between the North Carolina State University

Computer Science Department and staff therapists from the Division for Treatment and Education of Autistic and other Communications handicapped CHildren (TEACCH) at the University of North Carolina at Chapel Hill School of Medicine. The project was an exploration of whether virtual reality would be a suitable technology for enhancing the education of one such group of children.

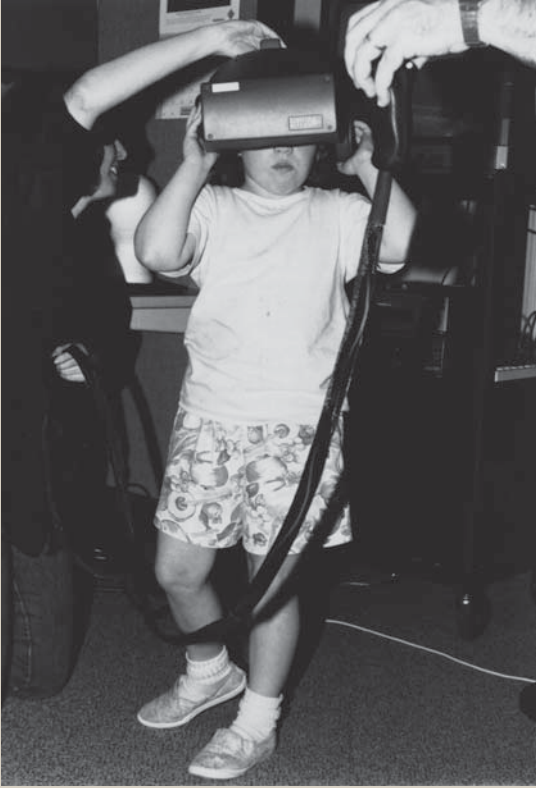


FIGURE 5-13

One problem with using virtual reality to teach children with autism is the fact that head-mounted displays are designed for adults. Here, you can see that not only is the HMD too large for the child but also heavy, as is evident from his using his hands to support the helmet.

Image courtesy Dorothy Strickland

For her project, Strickland chose to work with children with autism. Autism is a disability that causes children to be disconnected from their surroundings. They often display symptoms that include abnormal response to input stimuli, lack of human engagement, and inability to generalize between environments. People with autism often exhibit rigid, limited patterns of action and compulsive or ritualistic behaviors.

Application Description

It often takes longer to teach tasks to children with autism, and this can lead to real-world training involving dangerous environments resulting in considerable exposure to risk. Strickland hypothesized that VR would provide a safe learning environment for children with autism, and that a number of the special requirements for learning tools for these children can be met by virtual reality. This would be particularly beneficial for tasks that can be dangerous to learn in the real world, such as knowing what to do, and more importantly, what *not* to do, at street intersections.

When making an experience in any medium, it is always important to know (preferably early in the design/development process) who your audience is and what needs exist to specifically address this audience. The traits of autistic behavior constrain how and what information can be presented to people suffering from autism.

Strickland compiled a list of features that should be included in an environment for teaching children with autism [Strickland 1996]. Each of these features matches abilities of virtual reality, strengthening the possibility of the usefulness of VR for this task. Her list of features include:

- Controllable input stimuli
- Modification for generalization
- Safe-learning environment
- A primarily visual world
- Individualized treatment
- Learning with minimal human interaction
- Vestibular stimulation

The reasons for these features become more apparent when one considers the primary attributes of autistic behavior. Some common behaviors that make interacting with autistic people challenging include difficulty in focusing (easily distracted), difficulty in attending to more than one sense at a time, they often don't like things placed on their head, and (children especially) tend to get agitated when they are in unfamiliar surroundings. One thing they do seem to enjoy is "vestibular mismatch." This is when the sense of balance doesn't fit what the sense of sight is reporting. In the past, causing vestibular mismatch in students has even been used as a reward mechanism.

Of course, there are degrees to which individual people with autism can cope with the rest of the world. Their level of functioning is broadly classified as being low-, mid- or high-functioning. Low-functioning people with autism are not able to perform verbal communication. High-functioning people with autism are able to perform complex tasks well enough to hold a job. Mid-level function people with autism can do rudimentary verbal communication, but are not able to cope in most situations.

Strickland's initial work has been with mid-level functioning children with autism. This group needs more educational help than the high-functioning group. The level of communication with mid-level individuals with autism is limited to a few words. In this case, they cannot speak or understand normal sentences, but they can understand a few simple verbal instructions. Even this limited vocabulary of a few words and ability to follow simple directions makes it more straightforward to work with mid-level than low-functioning children with autism.

For months prior to designing the application, Strickland visited a classroom to study the children's behaviors and get ideas for how to best design the experiment and how to create an environment amenable to the children. One measure that was used was to have the toys that were familiar to the children at the lab environment. Since many autistic children tend to get agitated when in unfamiliar surroundings, or when not allowed to follow their normal routine, providing familiar toys to the lab environment and maintaining a consistent schedule helped to alleviate some agitation.

In order to build a VR environment based on her list of important features, Strickland created a simple scene with a street, a sidewalk, and texture-mapped buildings. She chose not to include any people or animals, or any objects in the sky. An important consideration when dealing with children with autism is to avoid unwanted distractions, so the only nonstationary object in the scene was a single car that passed by periodically.

The only output of the system was visual. Instructors gave verbal instructions to the subjects, but the computer simulation generated no sounds. Because children with autism often get agitated when required to attend to more

than one sense at a time, the decision was made to focus on only the visual elements of the task. The application authors created a low-contrast environment using gray as the predominant color. The cars were made bright red and blue, the two colors recognized by the selected subjects in pre-VR cognitive testing. These colors also provide high contrast against a gray background.

Each trial would begin with the subject located in one of a selection of different places to allow the child to experience different scenery at the street crossing position.

Since the children were minimally verbal, it was difficult to give the children instructions, and likewise difficult for them respond to the instructors. To overcome this difficulty, the parents aided by physically helping the children carry out the instructions the first time. When successful, they rewarded the children and repeated the process in order to help the child associate the words with the task.

Strickland altered the scene between sessions to accommodate each subject's response patterns. Features that she could control from trial to trial include the speed of the car and the particular intersection at which they stood. A small virtual downtown was modeled, and the child could be placed at a variety of locations such as the middle of the street, in front of a gas station, across from stores, or at the corner of a two-way intersection. Strickland edited the world file by hand between trials to setup each particular scenario.

The First Experiment

Strickland developed an experiment to test whether a child with autism would understand the virtual world presented in an HMD, could learn a task in VR, and whether such learning would generalize to the real world. Two subjects diagnosed as being autistic were chosen for this study, a 7-year-old girl and a 9-year-old boy. Both were classified as mid-level functioning. Each child was exposed to the VR environment 40 times over a 6-week period.

The first task was for the child to learn to recognize an object. In this application, the object of primary importance was a car on the street. Once they were able to accomplish the recognition task, the child was instructed to find the object in the scene, walk to it, and stop. In addition to the car, there was also a stop sign located in the scene that the child had to find and walk to.

**FIGURE 5-14**

A simple virtual world was modeled that had some simple buildings and vehicles.

Image courtesy Dorothy Strickland

Because children with autism are often not skilled at knowing the most important things to attend to, it was important to provide an environment that promoted attention to the appropriate task. Also, there were no human representations in the virtual world. To avoid the distraction of room noises, and the sounds of the hardware, they placed padded inserts in the headphones of the head-mounted display. The occlusive nature of the HMD helped prevent visual distractions. The students could not see the other people while they were in the helmet. This feature of HMDs is one of the major advantages over a projection VR environment for this type of application.

No exposure was over 5 minutes. A trait common among children with autism is an aversion to having things on their heads, so there was concern over whether or not the children would even accept wearing the head-mounted display. Preconditioning was used to increase the likelihood that the children would wear the HMD. The boy practiced with a football helmet and the girl a riding helmet prior to trying the HMD. Also, the children's siblings were present as an aid in getting the children to accept the HMD. The subjects were more accepting of the HMD when they saw their siblings using and enjoying it. The girl subject immediately accepted the helmet. The boy, on

**FIGURE 5-15**

Children were able to see the virtual world on a computer monitor before they experienced the application using the HMD.

Image courtesy Dorothy Strickland

the other hand, didn't take the helmet until the second session. Interestingly, once he donned the helmet, he became calmer.

Since the virtual world was also displayed on a monitor in the test area, the children were able to see the car and virtual world on the monitor when they were not in the HMD. Interestingly, even though the children could not see the flat monitor once they had donned the HMD, both insisted that the monitor display be present before accepting the HMD.

Once they accepted the helmet, the children were able to learn to do the tasks. They would turn their bodies physically as appropriate to track the movement of the cars.

However, each child seemed to have a different acceptance of the idea of being immersed in the illusion. For example, when asked to point to the stop sign the boy would point to the front of the HMD (i.e., the location of the physical image of the sign), whereas the girl would walk to the stop sign and point to a position in the virtual world, although she too sometimes pointed to the front of the HMD. The boy was familiar with conventional PCs, so this may have contributed to his overlapping the immersive and real worlds. For this reason, they were instructed to walk to the object rather than point at it.

There was evidence of generalization in that the children responded similarly to three different street scenes, but Strickland knew that further study was necessary to determine if the children were generalizing across environments. Overall, the fact that the children seemed to accept and respond to the virtual worlds shows that VR might be a tool that can aid in the treatment of autism.

VR System

The virtual world was created using the Division dVS development environment and the *dVISE* authoring and simulation program software provided with the system. Development of the experiment was done over the course of a month of evenings by Strickland, on her own time. Use of the hardware was donated by Division Inc. Strickland was allowed to use equipment at Division's sales office in North Carolina during off-hours.

The VR system used in this application was centered on a ProVision 100™ integrated VR system provided by Division Inc. The system included an integrated Pixel-Plane hardware board for rendering the imagery, and an HMD. The HMD was a Division dVisor that provides 345 by 259 pixels of resolution with approximately 105 degrees horizontal FOV. The images were displayed stereoscopically, and adjusted for each child's interocular distance. The HMD weighed about 8 pounds, which was heavy for the children. Near the end of the sessions, the children were supporting the weight with their hands. A Polhemus FasTrak™ magnetic position tracker tracked the HMD in full 6-DOF.

In addition to the isolation benefit derived from wearing an HMD, the head-based type of visual display device is

also less expensive and far more portable than projection VR displays. For all these reasons, HMDs were considered by Strickland to be the best choice for this particular application.

The Follow-up Study

A 6-month follow-up study was funded by an SBIR (Small Business Initiative Research) Phase I grant from the National Institute for Health (NIH). This grant was given to test whether students with autism who learn in the virtual environment will transfer that information to the real world, whether the training can be done on low-cost machines, and whether such a system is commercially viable.

The specifics of the experimental test were to analyze how well children with autism accept the technology, how well they learn concepts in the technology, and whether what is learned will carry over to the physical world.

In this study the specific task was "object identification." Strickland used subjects who were already participating in a cooking class, and who had mastered the ability to use a standard computer mouse. The task was to learn the names of different utensils used in cooking, such as a spatula. Objects used in the experiment were ones that had not yet been learned. The student identified objects by "picking out" the object in the HMD using a standard 2D desktop mouse.

Strickland reports that the results were impressive. Three objects were presented during 5-minute sessions. All of the subjects were able to learn the objects and identify them quickly in just two or three sessions. Once the students accomplished the object identification in the HMD they were tested in the real world with the real, physical objects. If they were not able to identify the real object, then they were retrained in the HMD. One to two weeks later, the students were retested with the real-world objects, and the statistical measure of their ability to identify them was better than chance.

To reduce costs both in development time and hardware expense, the object models were extremely simplified. By representing objects as simple 2D-texture images mapped onto a 2D polygon, the laborious task of object modeling and the computationally expensive task of rendering objects both become feasible in a small scale project.

This study was done without any user tracking, so no hardware beyond the computer and HMD was necessary. In fact, without tracking, the HMD was only beneficial because of its occlusive nature and the reduced distraction it provided the user. Strickland states that a flat screen with a hood to block distractions may have worked almost as well. However, once the children were used to the HMD, they liked it enough that it became a reward for the learning endeavor. The specific HMDs used in this study were Virtuality arcade-VR displays rented from n-Vision for the 3-month duration of the trials.

A Related Project

A related project was *Project S.O.L.V.E.* (*Studies on Learning in Virtual Environments*), a joint effort between the Education and Outreach group at the National Center for Supercomputing Applications and the Transition Research Institute at the University of Illinois at Urbana-Champaign. The goal of *Project S.O.L.V.E.* was to research how VR can be applied in training and education. One specific skill they also chose to investigate using VR is children crossing streets. As part of this study, they chose to compare children with various disabilities to those without. The disabilities included autism and brain injury, among others.

The NCSA street-crossing simulation was created for the *CAVE* environment. The *CAVE* was chosen for a variety of reasons. Foremost was that the application was developed at NCSA, where a *CAVE* was readily available. However, use of the *CAVE* also provided an interesting contrast to other efforts at educational tasks. The *CAVE* can be more of a group experience, and at the very least does not isolate the subject from the instructor. The NCSA *CAVE* trades off loss of field of regard for wider field of view.

This application provides multiple street-crossing scenarios, including crossings protected by traffic signals, crossings protected by stop signs, and unprotected crossings. The *S.O.L.V.E.* simulation is designed to be a more realistic representation of the practice environment. Thus it is more complex than Strickland's worlds with multiple cars, buses, and a very rich environment with buildings, trees, and other details such as fire hydrants and street-lamps. One of the corners is a model of an actual intersection in Urbana, Illinois, near the school of the children involved in the study, which would thus be familiar to some of them.



FIGURE 5-16

A student stands in the CAVE ready to cross the street after the truck passes. An instructor watches from within the CAVE.

In this version of the street-intersection training task, the students were asked to go ahead and cross the virtual street at the appropriate time based on traffic and the type of traffic signs. To move the distance necessary to cross a street requires some travel paradigm other than physical locomotion as the *CAVE* is smaller than a street intersection.

Since the audience was young children a simple method of virtual travel was necessary. The experimental task only requires that the subject move forward to cross a street, so travel is controlled by pressing a button to move forward, and releasing the button to stop. This method of travel allows the student to approach the street, stop, look both ways, and when safe to do so, continue across the street. To familiarize subjects with the VR environment and practice the travel control, they were given a training session in a special pretask application. This task involved standing in a colorful field with a flower and tree ahead of the subject. They were instructed to approach the flower and stop. Stopping at the appropriate time rewards the child with sounds of applause and cheering.



FIGURE 5-17

Since many of the students in the study were from the same school, a corner near the school in Urbana, Illinois, was modeled for the study. Many students recognized the corner.

Preliminary Results

The project collected experimental data on 81 schoolchildren to see whether they could follow directions and safely complete the task and generalize between virtual environments. Half of the subjects had various disabilities. The data was analyzed at the Transition Research Institute. Analysis showed positive effects from training in the *CAVE*.

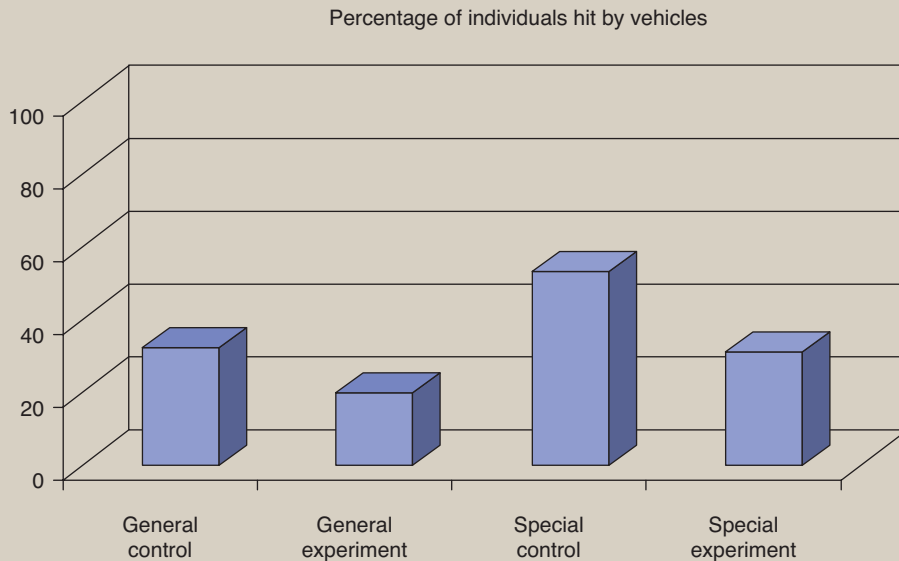
Another interesting preliminary finding indicates differences in difficulty based on the type of intersection. Based on the number of “virtual” accidents, the four-way stop intersection was found to be the most complicated intersection for children to maneuver, versus two-way stop and stop-light controlled intersections.

In addition to the findings regarding children and street crossing tasks, the researchers also learned an important lesson about VR application design. In an attempt to prevent the subject from walking laterally off the sidewalk, they implemented a scheme to move the world with the subject's head. The inadvertent result of this was to eliminate the motion parallax depth cue. The resultant loss of motion parallax makes it harder for viewers to get an accurate sense of scale of the virtual world, plus makes it seem as though the world “swims around” whenever the viewer moves their head. As it turned out, subjects don't really do much physical movement anyway, so it wasn't really necessary to do anything to keep them from walking sideways off the sidewalk.

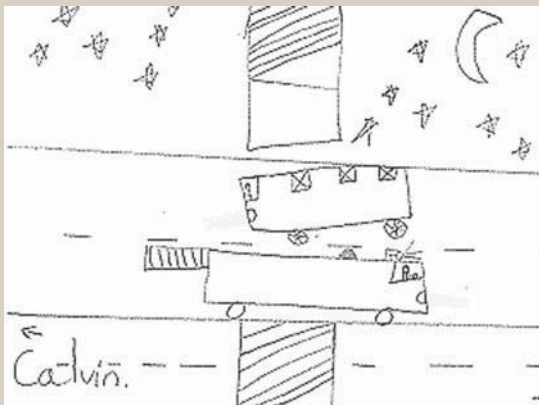
Another design issue involved the 3D glasses. The NCSA *CAVE* uses the Crystal Eyes shuttering glasses, which were too large for most of the children involved in the study. To accommodate the smaller size of the children's heads, the researchers attached the glasses to a hat that the children were able to wear more comfortably. After one of the parents mentioned that they had a hygiene concern regarding the sharing of the hat, they went back to using the ill-fitting glasses.

Conclusion

These projects laid groundwork for future work in using virtual reality to teach children potentially dangerous tasks in the safety of a virtual world. The *S.O.L.V.E* project expanded toward training of potentially dangerous scenarios to a wider base of children, and relied on the expectation that what is expensive VR today will be affordable in the not too distant future. Strickland, on the other hand, looked at getting

**FIGURE 5-18**

Researchers from the Transition Institute at the University of Illinois studied the participants in the experiment. This chart shows the percentage of children hit by virtual vehicles in each experimental group.

**FIGURE 5-19**

This picture drawn by a fifth-grade special education student after his experience crossing the street in the CAVE shows one way in which the visit can be augmented through post-visit activities.

low-cost systems into use as soon as possible, and expanding the potential user base to help train low-functioning (nonverbal) children with autism, by using EEG monitoring to determine the response of the subject.

The expense of the CAVE makes such applications impractical for daily use by a majority of students. However, the use of special field trips to interact with applications specially designed for the students' skill levels can be a beneficial learning tool. Such experiences can be augmented by preparatory instruction about what they will do/see at the CAVE, and post-visit discussions about the experience. The education continues after the time spent at the VR facility ends. Discussions and other activities continued back in the classroom build on the experience in the CAVE.

CyberEdge Journal awarded Strickland its product application award for 1995, stating it to be an example of "finding a new way to use a powerful technology to help the least empowered."

CASE STUDY 5.4: ULTRASOUND DIAGNOSIS USING AUGMENTED REALITY

Executive Summary

Visualizing complex three-dimensional structures using tools capable of only two dimensional representations can be a difficult task for many people. Yet many professions require this very feat, and eventually the practitioners learn how to create 3D mental models given the 2D display as presented by 2D tools. Of these professions, perhaps the medical field offers the greatest challenge to coping with this task because new technologies and new uses for existing technologies are frequently introduced. Medical procedures often have the requirement of visualizing structures internal to the body, with only limited access to seeing the 3D structure directly.

Aiding doctors in overcoming this challenge is one way that researchers at the University of North Carolina at Chapel Hill (UNC) Computer Science department are using their programming skills. Fred Brooks, professor and founding department head of Computer Science at UNC, believes that computer scientists should do their work with an eye toward what is required to solve real-world needs. The CS department was interested in researching augmented reality (AR)—a technology related to virtual reality that combines the senses of the real world with computer-enhanced imagery. Building on a history of collaboration with members of the medical profession, they chose to work with doctors to create a system with the goal to “develop and operate a system that allows a physician to see directly inside a patient.” Doctors use a variety of imaging technologies to help them visualize the internals of the human body, including x-ray, CAT scans, MRI scans, and ultrasound scans. The UNC team chose to work with ultrasound scanning technology for three reasons: it is fast, noninvasive, and does not use harmful radiation. An AR ultrasound system could be useful to doctors to “see in” patients during procedures such as obstetrics exams, cardiology exams, and diagnostic procedures such as needle-guided biopsies.

Ultrasound is widely used for procedures such as breast biopsies, removing tissue lesions, and localizing lesions prior to biopsy and cyst aspiration. Using ultrasound scans in these operations is “difficult to learn and perform.” It requires good hand-eye coordination as well as 3D visualization



FIGURE 5-20

Prof. Henry Fuchs watches as Dr. Etta Pisano performs a simple visualization procedure using an augmented reality visualization system.

Image courtesy University of North Carolina

skills. Prior to using an augmented reality display, physicians performing such operations had at their disposal only a “non-registered two-dimensional ultrasound image (and perhaps prescan 2D medical imagery such as x-ray images) to assist her in the inherently three-dimensional task of guiding a needle to a biopsy target.” [State, Livingston, et al.]

Challenges of Augmented Reality

Although augmented reality was not an entirely new technology (Sutherland’s original head-mounted display allowed the viewer to simultaneously see the real world and virtual imagery), AR application development and research had a very limited scope. Besides the development in the military of head-up displays, few VR developers have shown much interest in the technology until more recently.

Part of the challenge for this project was to find out what was possible with then-current technology, and then set out to try to solve the problems that prevented the system from being useful as a medical tool. The project was begun by Professor Henry Fuchs in 1992. As the first medical procedure to explore with AR, he chose obstetric fetal exams

as a good starting point. For reasons discussed below, the research team later changed their focus to other medical operations. A major focus was on the use of AR for “ultrasound-guided needle biopsy of the breast” operations.

Although virtual reality benefits from rendering the virtual world in as many senses as practical, augmented reality benefits from the circumstance that the user is already immersed in the real—physical—world. The physical world itself provides feedback for all the senses, including vision, so an AR system need only focus on particular senses to help convey additional information about the real world. For this reason, augmented reality systems often focus on representing, rendering, and displaying only visual information. Such is the case with the UNC *Ultrasound Visualization* project.

One thing they knew from the outset was that the artificial world would need to be rendered and displayed such that viewers “believe that the synthetic objects exist in the [real] environment.” [Jacobs et al.] To realize this goal would require advances in registration, latency, resolution, and information representation. Problems on all fronts were encountered, and to a great extent reduced during the course of this (ongoing) project.

Registration

Registering the real and virtual worlds, such that objects coincident in both worlds appear to be in exactly the same place at all times, is one of the most critical problems for augmented reality. Registration affects the mental immersiveness/believability of the illusion, and in this case, usefulness. If the artificial objects are not properly registered, it is very difficult for the viewer to determine where those objects are supposed to be positioned with respect to the real world. Solving or reducing registration errors is crucial.

Both spatial registration and temporal registration are important to merging the two realities. Spatial registration is fairly obvious, but if objects are not aligned temporally, then when the user turns their head, they will see the real-world change, followed by the virtual-world change a few instances later. This delay makes it apparent which objects are real, and which are not. The delay between the real-world view and the virtual world objects is referred to as *relative latency*. In his research at UNC, Rich Holloway

found that about 1 ms of latency in tracking results in up to 1 mm of registration error at arm’s length, ignoring positional error and assuming a peak angular velocity of 100 degrees/sec for the user’s head. [Holloway 1995].

Latency

Latency is a common problem for both virtual reality and augmented reality, but usually more so for the latter. Latency can enter a system in many different ways. From work done by Wloka [Wloka 1995], Mine [Mine 1993], and Taylor [Taylor et al. 1995], the UNC research team compiled a list of six ways in which latency is introduced to an AR or VR system:

- “Off-host delay”—the delay between the actual event, and the *sensing* of that event by the computational system.
- “Computational delay”—the delay caused by doing computations on the incoming signal to produce a world-state.
- “Rendering delay”—the time required to render the world imagery.
- “Display delay”—the time between creating the rendered image, and the display of that image. (In visual rendering and display, this is generally the time spent waiting for the next screen refresh to occur, and presenting the new image at the beginning of the next display scan).
- “Synchronization delay”—the time waiting for some other crucial data path to catch up.
- “Frame-rate-induced delay”—the amount of time since the current image was first presented. For example, in the visual system the image is already latent when first placed on the screen, but as time passes waiting for the next “frame,” the current image continues to age and become increasingly delayed from the incoming data.

Using the above terms, the relative latency is a combination of “off-host delay,” “computational delay,” “synchronization delay,” and in some cases “rendering delay.” Rendering delay is particularly problematic when the optical see-through style of HMD is used because the “rendering”

of the synthetic imagery cannot be fast enough to match the real world (which does not exhibit a rendering delay). The other forms of delay will of course negatively affect the overall latency of the images presented, but do not affect the relative latency because all the data go through these steps together.

Resolution

The degree to which one can precisely sense the environment plays a large role in how we relate to it. The greater the amount of detail we perceive adds to both the realism of the image and its usefulness. If a visual image is so blurry that we would be considered legally blind, then we will have less ability to perform functions than if we had simulated 20/20 vision.

The need for greater resolution is even more crucial when the functions being performed have real-world consequences. This is obviously true for the medical procedures where fine hand-eye control is necessary to perform the operation.

Visual resolution can only be as good as the display device supports. When using the video see-through style of AR display, even perception of the real-world visual scene is limited by the HMD. In the case of needle biopsy operations, the 22-gauge needle that is typically used is less than the width of a single pixel in the VR-4 HMD used for many of the early UNC experiments (about 0.7 pixels thick). A 3mm breast lesion would be about 3×3 pixels in size on the VR-4 display.

The solution to this was fairly obvious—use higher-resolution displays and cameras. At the time (1996), the UNC team did not concern itself with trying to solve the problem of video resolution themselves, because it would be reduced by natural advances in video technology. For their analysis, they have instead chosen to use thicker needles. Of course, this limited their testing to training dummies, since they cannot use the thicker needles on real subjects. They have subsequently upgraded to a higher-resolution HMD, allowing more realistic testing.

Interface with the Augmented World

AR systems are constrained by how the user can interface with the world. One constraint for most AR systems is that the user must occupy the same physical location as what

is being represented in their display. While limiting the options for how the user interfaces with the “virtual” world, it also makes the task much simpler. For instance, the only method of travel that makes any sense is *physical locomotion*. If the user doesn’t move in the real world, then they shouldn’t move in the augmented world. Also, many of the world-manipulation operations are done exactly as they would be done in the absence of an AR system. For example, to move the position of the ultrasound-augmented sensory input representation the user simply moves the physical ultrasound probe as they would under normal circumstances.

An example of a scenario that is not subject to these constraints is when AR is combined with telepresence. Because telepresence provides a remote position for the human participant, their interactions with the real world are mediated. Thus other VR interaction paradigms are possible. Also, the constraint requiring the participant to occupy the same physical location is replaced by requiring the participant’s surrogate senses (robot) to occupy that space. So a point-to-fly method of travel could be implemented provided the sensory-surrogate can perform the maneuver in whatever real-world space it might be (e.g., underground, in space, or on land).

One freedom is that there is less need to attempt to represent a “photorealistic” picture of the augmenting information. In general, when participants are using an AR system, they want to see something not directly viewable, and therefore not directly able to be represented. Thus, the challenge is to find ways to represent what is behind the walls, or beneath the skin, in an intuitive manner, despite the fact that seeing directly through solid objects is not something with which the users will have had a lot of experience (although doctors will of course have experience in seeing 2D representations of 3D structures, but this information is gathered “off-line” and is not viewed in registration with the solid object). The UNC research team discovered many features of good and bad representations as the project progressed.

Representation of the Augmented World

A significant factor in whether an AR system will be considered useful to a doctor is how well the information presented to them is represented in a way that makes it easier

**FIGURE 5-21**

Visualizing ultrasound images without adding proper cues that indicate placement causes the images to be ambiguously floating in space.

Image courtesy University of North Carolina

for them to grasp the nature of the situation. In basic terms, doctors are interfacing with three-dimensional objects using two-dimensional sensing devices. By moving these 2D sensing devices around, they can scan the entire 3D structure, but it can still be difficult to perceive the overall nature of the object. A goal of this project has been to find effective representations for these objects within the constraints of the scanning technology.

Floating Slices

At the beginning of the project (when using a fetus examination as the prototype task), the first style of representation was to simply place 2D ultrasound slices of the developing child over the image of the woman. This technique was not very good at all. It was nearly impossible for the doctor to determine the position within the body where the slices were located.

The Pit (hole)

A direct result of trying to solve the problem of the “floating slices” representation, the use of a *pit* to represent a hole in the object (i.e., the patient) was found to be a much better representation. The pit gives location context to the scanned data, and allows the data to be rendered with the proper occlusion cues.

The pit appears as the inner-surface walls, floor, and rim of a hole in the object. Other forms of data are then rendered to appear within the pit, such as 2D ultrasound slices. Representation of material that is inside the body but not within the pit is occluded by the pit walls. Later representational experiments removed the inner-pit surfaces, leaving just a hole through which the doctor can explore the inner cavity of the patient. The plain hole allows the viewer to look obliquely through the hole, which is useful if there is some benefit from not “removing” too much of the skin surface, yet still wanting to be able to see underneath a larger area of the skin.

For best results, rendering a pit or hole requires that the edge of the pit/hole match the surface into which one is peering. Without this edge matching, the problem of the floating slices begins to resurface as it becomes difficult to determine how the pit/hole matches to the body, and therefore how the objects within the pit/hole are positioned. In order to properly render these edges, the system must know the shape of the surface, thus requiring more sensing technology.

The first representations placed inside the hole were the same floating slices as in the earlier versions. However, any type of representation can be placed within the hole, including isosurface and volumetric rendering techniques. Two other representations implemented by the research team thus far are a sphere to represent the target tissue (a cyst or tumor), and a representation of the biopsy needle (which must be registered with the actual needle).

Radar Slices

Ultrasound data is “accurate” only for the exact moment of each scan, because tissue within the body (especially a developing child) does not remain motionless. Thus, continuing to display all previously scanned data can misrepresent the hidden tissue. To address this, a “3D radar” idiom was created in which older scan slices are faded out over time, eventually disappearing altogether. This is very reminiscent to how radar displays represented information about echoes from the environment—as the probe scans in a certain direction, a bright blip is produced on the screen. When the probe moves away, the blip begins to fade.

The rate at which old slices are eliminated can be selected by the user. Choosing the fade rate is influenced by the

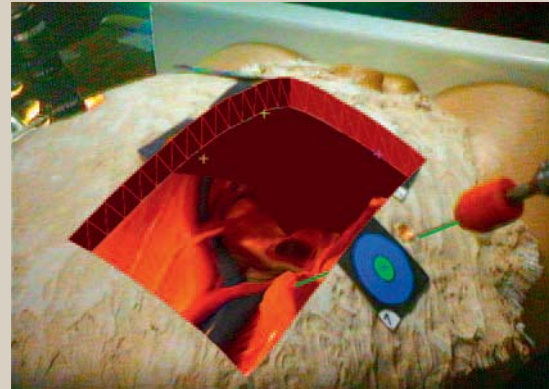
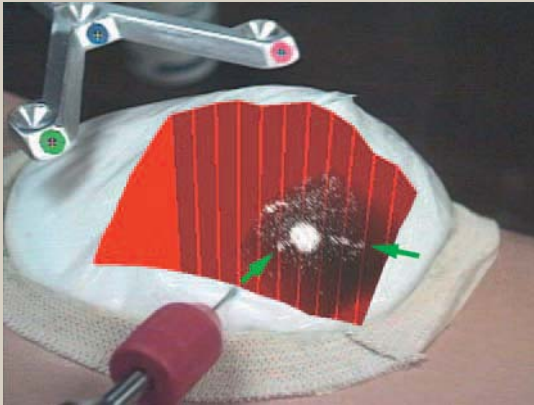


FIGURE 5-22

Two methods of the “pit” representation. On the left, the pit extends several inches, with a floor on the bottom. On the right, the walls of the pit are only about an inch deep, showing the rest of the body cavity beneath.

Image courtesy University of North Carolina

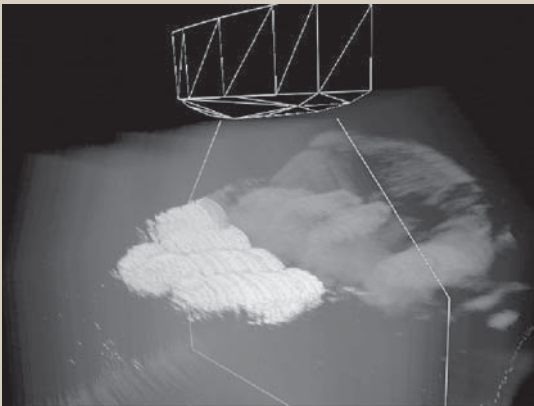


FIGURE 5-23

As the ultrasound scanner moves, old scan images begin to fade away, indicating the degrading confidence in the data.

Image courtesy University of North Carolina

amount of clutter versus the need for lots of data, and by how much the underlying tissue can be expected to move. For example, a developing child is likely to move a lot more than a lesion within the breast.

Depth Cues

Depth cues are always important in rendering a believable visual scene. In augmented reality, they become even more important as a means by which the synthetic imagery is represented to appear as though it occupies the same space as the real world. Four visual depth cues that are particularly important for enabling this effect are head motion parallax, stereoscopic imaging, proper occlusion, and lighting and shadow effects.

Head Motion Parallax

Head motion parallax is treated no differently than with typical VR display systems, and is vitally important to creating a world that responds appropriately to user movement. An added concern for AR display systems is that the synthetic and real worlds should respond the same to any user movement of the head. This concern is another manifestation of the registration problem.

Stereoscopic Imaging

Physicians typically work in small nearby working volumes. This situation is very conducive for using the depth cues provided by stereoscopic visuals. The stereoscopic depth cue is needed to help the physicians work, and was thought to be required for acceptable performance.

Occlusion

Nearer objects occluding objects that are farther away is the strongest of all depth cues. Therefore, appropriate merging of synthetic objects into the real world must avoid improper object occlusion at all costs, or else the virtual imagery will not seem to be of the same world. In some cases, this may simply be accomplished by preventing synthetic objects from moving to locations where they are likely to be occluded incorrectly. However, especially when working in close quarters, these circumstances cannot always be avoided. Therefore, other techniques must be employed to produce a proper image.

There are two situations that need to be addressed: real objects in front of virtual objects should result in the virtual object being (perhaps partially) occluded by the real object, and similarly, virtual objects need to be able to hide real objects in the scene. These may seem to be the same issue, but the latter requires the use of the video see-through method of AR display, whereas the former can be accomplished with any AR display.

In either case, proper occlusion-rendering requires knowing the positions of the real and virtual objects that might be involved so the rendering system can determine where to render the synthetic objects. This means that any real-world objects that might occlude virtual objects need to be tracked or kept out of the way.

In addition to tracking the real-world objects, the system must know the shape of those objects. Objects for which the shape cannot be known *a priori* must be scanned just prior to the procedure. In particular, the shape of the body region that will be involved must be determined. Knowing this shape is particularly important for aligning the edge of the virtual hole with the surface of the body. If the edge of the pit doesn't match the body shape, then the parallax from head motion will result in the pit moving slightly separately from the body and thus not appear to be attached to it. To get the shape of the body, it is prescanned by a mechanical tracker just prior to the procedure. Once scanned, it is assumed that the patient does not move until the procedure is completed. This assumption is of course an approximation of what will happen, but according to developer Mark Livingston, it is a surprisingly "acceptable approximation that does not cause significant



FIGURE 5-24

Images of the probe, pit, and patient are properly overlain, providing correct occlusion cues, but the doctor's finger is not occluded properly.

Image courtesy University of North Carolina

error compared to other problems." He continues: "Until we have accurate head tracking, it doesn't pay to worry about accurate surface tracking."

To properly render virtual and real objects together in the same scene, the developers take advantage of the z-buffer technique² common to most computer graphics rendering hardware. This is done by rendering the depth information of the real-world objects, but not their physical appearance. This "trick" then allows the virtual scene to be rendered normally, and any information from virtual objects that would be rendered behind other objects, including the z-rendered real objects, will not appear in the image. This technique works for both the optical and video methods of see-through HMD displays.

Proper occlusion rendering allows the scene to include virtual objects that visually behave as expected. For example, a virtual object can penetrate the skin but will only be seen if it happens to be within the pit. Some real-world objects

² The z-buffer rendering technique is described in Computer Graphics: Principles and Practice [Foley, VanDam, Feiner, and Hughes], Computer Graphics [Hearn and Baker], among others.



FIGURE 5-25

Lighting techniques are used to create more photorealistic representations of virtual objects. Here, a computer-generated teapot has a realistic reflection map to enhance the appearance of the object. Fiducial markers in the world allow the objects in the world as well as the viewing camera to be accurately tracked. Image courtesy University of North Carolina

are too difficult to track effectively (such as the doctor's hands) and thus will not properly occlude the synthetic objects. The best solution in this case is to reduce the need for the hands to come between their eyes and the virtual objects. Of course, this cannot always be accomplished.

Lighting and Shadows

Lighting effects (including shadows) are also useful in helping a viewer determine the relative location of objects. Of course, when dealing with the representation of non-real, in fact, ethereal images, lack of shadows and interaction with lights is less important than when photorealistic imagery is the goal. The UNC research efforts toward producing a medical visualization system have thus been able to avoid addressing lighting effects. A separate effort at UNC has done research on photorealistic rendering of augmented reality scenes [State, Hirota, et al.].

Technology Enhancement for Useful AR Systems

Just as important as the efforts to produce a usable VR tool are, producing an augmented reality tool requires not just

the development of the interface and the representation, but also requires the pursuit of advances in the technology. That is, there were certain problems not related to the specific task of medical visualization that needed to be solved to bring their system to fruition.

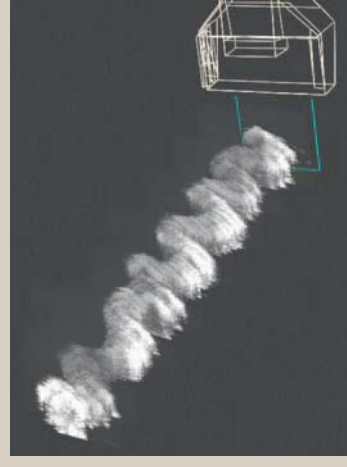
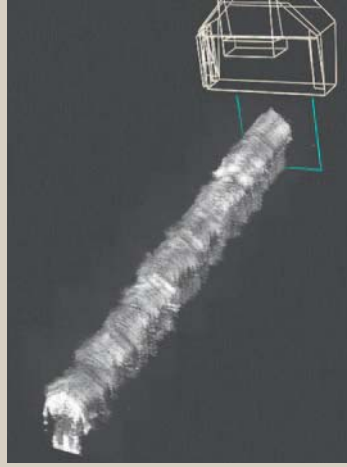
One goal of the UNC project was to uncover what shortcomings and needs existed in technology that precluded the creation of a usable augmented reality application. These shortcomings were not so great as to cause the developers to give up in their pursuit to develop AR applications. Rather, it required parallel development efforts to solve, reduce, or avoid problems that were showstoppers in the original implementation.

One such effort was to develop a video that would demonstrate some high standards to which they could aspire. This attempted "gold standard" would be produced using as many of the same technologies as the struggling AR system as possible. The video was produced by recording all the incoming data along with time-code information such that it could be rendered and edited together in non-real-time. By freeing themselves from the constraints of insufficient computational power and delays in the data gathering and display systems (resulting in the total elimination of all types of latency), they were able to present a view of the possibilities of an AR application in the absence of technological bottlenecks. The result of this effort was not quite as good as they hoped for, and they deemed the video to be a "brass standard" that still represented something to strive for.

To make the "live" system function at a more acceptable level, the UNC research team created a system that was more usable by confronting three main issues: addressing a simpler problem, reducing relative latency, and enhancing tracker performance.

A Simpler Problem

After their initial work with visualizing a child developing within the womb, one of the main problems that made the AR system less than optimal was the fact that the baby frequently moved during the exam, making it difficult if not impossible to use the system in a meaningful way. Since other medical procedures that rely on ultrasound information examine tissue that doesn't move around quite so much, it was decided to address those procedures first.

**FIGURE 5-26**

Without paying attention to relative delays between inputs, the path followed by the scanner can cause the data to be similarly skewed. The image on the right shows the more accurate representation obtained when information is properly aligned.
 Image courtesy University of North Carolina

The research could return to the fetus examination later, once some of the other problems had been resolved technologies improved.

Relative Latency

Relative latency between incoming data streams leads to two problems: faulty temporal registration and incorrect recreation of scanned objects. When the ultrasound data is not properly aligned in terms of time, objects that are physically straight can end up being stored internally and represented in a shape highly affected by the scanning. For instance, if a zigzag or curving pattern followed during the scan is used, a straight object will look zigzagged or curved.

The two primary ways of addressing and reducing relative latency are to use time stamping and predictive filtering to temporally align data input streams, and also to delay inputs of lower latency to match those of higher latency. By using the video see-through method of AR visual display, the system can delay the real-world imagery to match it with the computer-rendered world. This temporal matching of images comes at the expense of delayed response of the real world to user movements. Thus, it is

imperative that the overall delay not be very long, or the user can experience nausea and/or other problems.

Time-stamping the ultrasound data at the moment when the scan was made and recording the location of the probe across time allows the resultant data to be positioned at the spot where the probe was located when the scan was made rather than where the probe is at the time it is displayed. This means that the user will see a delay between the probe's movement and the appearance of scan information, but the representation of the world will be more accurate.

Tracking Performance

Good tracking is important for a virtual reality system to give the user the opportunity to suspend disbelief and mentally immerse themselves into the virtual world. For an augmented reality system, it is vital. The error in registration between real and synthetic worlds caused by bad tracking spoils the illusion that the two worlds coexist.

As described in *Understanding Virtual Reality* [Sherman and Craig], different tracking technologies offer different abilities and constraints.

Magnetic tracking systems are nice because they are not very encumbering, allowing the tracked object (e.g., head, ultrasound probe) to be freely moved about the space. However, these systems suffer from being not very precise or accurate in high-metal, high-RF environments. It is not possible to eliminate all the sources of distortion from the environment, because the list includes the ceiling, lights, floor, and the ultrasound scanner itself. Thus, if magnetic tracking is used, other ways must be integrated to reduce tracking error.

Mechanical tracking systems are nice because they are very accurate and precise, and have very low latency in reporting the current position. However, the mechanical linkages that connect the probe to the fixed reference point can encumber the user. Because of the benefits of mechanical tracking, if feasible, it would be ideal for as many of the objects as possible to be tracked mechanically. In some cases, this does not even add an extra burden to the system, because of linkages already in place. For example, the ultrasound probe is already linked to the ultrasound processing box, and adding the mechanical tracking to this only slightly reduces how the probe can be maneuvered. In general, though, mechanical tracking, while okay for many props and medical devices, is not good for tracking the head.

Optical tracking systems can also provide good precision, without the added movement constraints of the mechanical systems. However, it depends on a clear line of sight being maintained between the unit being tracked and the sensors. In a hospital setting, it is not unreasonable to expect that there will be little visual interference between the head of the doctor and the ceiling, so this is an acceptable alternative.

Videometric tracking is a feasible choice for tracking in any AR system using a video see-through HMD, as is the case for the UNC project. Videometric tracking is done by processing the incoming video stream, locating multiple landmarks, and using the location of these landmarks in the video field to calculate the position of the camera. The benefit of using a videometric system is that given enough landmarks, it can be very accurate, and when used in a system that already has a camera input, it doesn't add any encumbrances to the system. The major problem

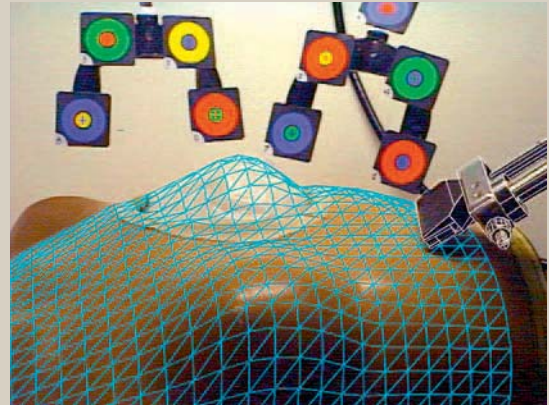


FIGURE 5-27

This image shows a wireframe representation of the body as it is scanned by a mechanical arm. A collection of colored fiducial landmarks are used to help register the real and virtual visual images.

Image courtesy University of North Carolina

with this type of tracking is that it can be computationally expensive.

The UNC developers describe two techniques that they use to reduce the computational expense of tracking: using a known approximation for the tracked position to begin the process, and using easily identifiable fiducial landmarks for the vision system to make use of. The UNC system uses color-coded fiducial landmarks to enable quick identification of any particular landmark. They use two concentric circles, each colored by one of four bright colors, which allows them to have up to 12 unique landmarks.

A hybrid tracking system was ultimately chosen as the method of tracking the user's head. Implementing this system required the development of algorithms to integrate information from three separate tracking systems into a single data stream. The implementation details are described in a paper by Andrei State and colleagues [State, Hirota, et al.]. We will briefly summarize the technique and results.

The hybrid tracking system was chosen only for tracking the user's head. Because the simpler (and better) mechanical tracking method was feasible for the ultrasound probe, that

is how the probe was tracked. The hybrid head-tracking algorithm began by using the raw data from a magnetic tracking system, then using a static correction table (generated ahead of time using the mechanical system as the reference) to determine the amount of correction needed for many points in the working volume. This technique results in a decent approximation of the head's position, which can be fine tuned using the video information from the cameras.

In their research on tracking, the UNC team found that in a high-metal environment location errors of the magnetic tracking system could be up to 10 cm (i.e., huge), with orientation errors averaging 3.5 degrees (maximum of 6.6 degrees). The static correction table alone reduced maximum location errors by 80% (down to 2 cm), but only reduced the average orientation error by 40% (down to 2.1 degrees). The further enhancement from the videometric system reduced the errors to virtually nil.

The computational need of the videometric tracking approach can be reduced by basing the first-guess of the sensor position from the statically calibrated magnetic tracking information. Using a reasonable first-guess allows the image searching routines to know where to look for the fiducial landmarks. The location of the fiducial landmarks is preset by touching the mechanical tracker to each of them prior to beginning the operation.

It is sometimes difficult to keep many fiducial landmarks in view of the cameras. The operator cannot be concerned with this detail, so they must be strategically placed where they are likely to be within view of the camera. Three landmarks are all that are needed for "perfect" tracking. With two landmarks in view, then the camera's orientation can be completely determined. If only one landmark is in view, then two of the three camera orientation angles can be calculated. When no landmarks are visible, the system continues to use the latest known correction values.

Of course, better initial tracking reduces the necessity for more fiducials. After switching to an optical tracking system for tracking the doctor's head, the accuracy prior to correction from the videometric tracking was found to be good enough that a single fiducial reference point in view was sufficient to produce tracking results as good as the hybrid magnetic/videometric system with multiple fiducials.

Proper Rendering of Multiple Slices

The use of translucent slices results in loss of the ability to use the normal z-buffer method of fast rendering. Without the z-buffer, objects must be rendered in order from furthest to closest to the viewer. A technique for enabling objects to be quickly sorted is the binary space partitioning tree (BSP tree) algorithm. However, the BSP tree algorithm assumes objects are not added and deleted at a rapid pace as is the case of the fading radar slices. The application development team developed a method to use dual BSP trees to overcome this limitation as described in [Garrett et al.].

The AR System

As mentioned earlier, the UNC research team chose early on to use a video see-through HMD for the AR display device. In 1996, the HMD was a composite of a Virtual Research model VR-4 HMD with two Panasonic GP-KS102 cameras mounted on top, and a 3-pound counterweight added to the back. The VR-4 displays about 250 by 230 pixels per eye with a stereo image overlap of about 80% (in the range of the 50 mm working volume). Each of the cameras was equipped with a Cosmocar F1.8 12.5 mm lens. This camera configuration provides a 28-degree horizontal FOV. The primary concern in coming up with this configuration was to keep the optical distortion minimal. The cameras are mounted 64 mm apart to match the typical inter-pupillary distance of most people, and are oriented to have a fixed convergence of 4 degrees. Later, they upgraded the HMD to a Kaiser Electro-Optics ProView30 system with the same camera system attached.

Magnetic tracking was done with the Ascension Flock of Birds™ tracker. This system can operate at a sample rate of 103 Hz. Mechanical tracking for the ultrasound and ancillary operations were done with a Metrecom IND-01™ from FARO Technologies, Inc. The FARO tracker had a maximum sample rate of 27 Hz. The ultrasound scanner was a PIE Medical Scanner 200.

Later, the Ascension head tracker was replaced with the FlashPoint™ 5000 optical tracking system by Image Guided Technologies designed specifically for the medical market. The FlashPoint system was also used to replace tracking many of the instruments used in the procedures.

The mechanically linked (and still more accurate) FAROarm is still used for calibration, but most other tracking is handled by the FlashPoint 5000.

In 1996, the graphics and computational unit was a Silicon Graphics Onyx™ equipped with RE2™ graphics engines and a Sirius Video™ unit for real-time video integration. This has subsequently been upgraded to an Onyx with Infinite Reality™ graphics. In the prototype development for a fetal examination (done in 1992), a Pixel-Planes-5³ graphics system was used. The video unit can access two simultaneous video input channels into the frame buffer, main memory, or texture memory with some constraints, and different time delays involved for each path. The camera images are combined by a QD Technology QD-1110 and sent to the Silicon Graphics frame buffer. The ultrasound video is captured directly into main memory by the Sirius Video hardware and resampled by the CPU to produce 256×256 monochrome (8-bit) pixel images. The resampled images are then transferred to the Reality Engine (RE2) texture memory.

It would be too difficult to buffer the incoming camera video signals, and it would be highly undesirable to add any delay to the user's view of the real world, so this input stream was used as the timing-anchor reference. Fortunately, the camera video inputs were found to have the lowest latency of all the devices, making it easier to make this the baseline for measuring input latency. The Faro mechanical tracker and Ascension magnetic tracker were both found to lag behind the camera by 30 ms, and the ultrasound scanner lagged the camera by 250 ms (as measured at 10 Hz). The considerable lag in the ultrasound video was due to the path the data took to get from video to the computational memory, as well as the processing required converting it to the appropriate format.

The Silicon Graphics RE2 is capable of displaying to multiple devices. In the UNC AR system, three outputs are used. Two outputs are set to 640×480 -pixel VGA format (approximately the same resolution as NTSC television), and fed to the two HMD screens, and to separate monitors for others to watch. A third high-resolution image

(1280×1024) is fed to a computer monitor and used to display the user interface.

The resultant system is capable of sustaining a frame rate of 10 Hz both for the HMD display and the ultrasound image capture. At slower frame rates it can render up to 16 million ultrasound pixels—255 two-dimensional slices each with 256×256 pixels. To achieve 10 Hz, only up to about 20 slices were rendered at a time.

SYSTEM CHOICE

Display Paradigm

Perhaps the most significant choice made in designing the system for ultrasound diagnosis using AR was choosing the video see-through rather than the optical see-through method for merging the real and synthetic worlds. The benefits of using the video method are that it supports “proper occlusion relationships between real and virtual objects (provided that depth of real objects is known or can be determined),” there is better “ability to balance the brightness of real and synthetic imagery,” it allows the latencies between real and virtual imagery to be matched, and it makes it easier to integrate videometric tracking into the system [State, Livingston, et al.].

The four major disadvantages of the video method are the lowered resolution of the real world because of the cameras and displays, the spatial offset between the location of the camera and the user's eyes, the added weight of the cameras, and the limited depth of field of the small (“lipstick”) cameras. However, all of these disadvantages can (and likely will) be solved by improved technology. Even without the effort of the UNC team, cameras will become lighter and capable of higher resolutions. The use of some simple optics can adjust the incoming images to match those of the human observer, eliminating the spatial offset. In fact, the UNC research team has already developed an HMD to eliminate the eye offset [Fuchs et al.].

³The Pixel-Planes series of graphics hardware were developed as part of a research program at UNC.

Credits

A great number of people have worked on this and other research efforts at UNC to support this project. Henry Fuchs, professor of computer science at UNC, conceived the project in 1992 and has nurtured it from the beginning. Mike Bajura was the principal designer of the system featured in the 1992 paper [Bajura, et al.]. Andrei State has been the technical leader on the project since 1993. Mary Whitton is the project manager. UNC professor Stephen Pizer helped with the research.

Several graduate research assistants have been involved with the project, including Ryutarou Ohbuchi, Michael Bajura, David Chen, Hong Chen, Christopher Tector, Andrew Brandt, William Garrett, Gentaro Hirota, Mark Livingston, Paul Rademacher, Jessica Crawford, Jeremy Ackerman, and Michael North.

The UNC computer scientists also worked with several physicians in their effort to produce a tool designed for use by doctors, including Vern Katz MD, Nancy Chescheir MD, Ricardo Hahn MD, Mark Deutchman MD, Matt Mauro MD, David Casalino MD, Etta D. Pisano MD, Anthony A. Meyer MD, Matt Mauro MD, and Melanie Mintzer MD.

Conclusion

Overall, the UNC project for ultrasound diagnosis using AR proved to be quite successful. It was successful at both demonstrating the real possibility of this technology to be useful in the medical operating arena, and successful at unearthing and reducing the technological barriers of implementation.

In general, project leader Henry Fuchs finds that implementing an AR system for visualization has greater “demands than either virtual environments or scientific visualization individually.” This is due in part because, as Fuchs states, “Virtual environment and scientific visualization systems lack the error-emphasizing cues that a combined system provides” [State et al.].

There are a host of specific advances of technology that could significantly enhance the ability to produce a higher-quality AR system.

- Data-sensing hardware could include the capability of time-stamping the data stream to enable better predictive and corrective calculations.
- UNIX could be enhanced to behave in a real-time manner.⁴
- Standard ultrasound scanners scan at a thickness of 1 cm to 2 cm. While this is beneficial in many procedures, it does not work well with the AR system. The UNC team is striving to acquire a special thin-slice ultrasound scanner like the type used in cardiology exams.
- The weight of the HMD/camera unit must be decreased.
- The resolution of the cameras and HMD displays need to be significantly increased.
- The light sensitivity of the cameras needs to be increased.
- A method of scanning the patient’s body (and the doctor’s) to more faithfully produce the proper occlusion cues between the real and virtual objects must be developed.

Because the system could not be used by the primary doctor during a medical procedure, they equipped another doctor with the system. The second doctor would then observe the procedure “over-the-shoulder” to evaluate the system. The doctors report that the primary problems are insufficient resolution, overweight HMD, and difficulty in keeping enough fiducial landmarks in view to keep the trackers operating at maximum performance. They also reported some problems with the synchronization of input sources [State, Livingston, et al.].

⁴Silicon Graphics does offer a real-time variant of their IRIX flavor of the UNIX operating system.



FIGURE 5-28

*An artist's/developer's conception of the UNC AR system once the technology becomes less obtrusive.
Image courtesy University of North Carolina/artwork by Andrei State*



FIGURE 5-29

*Doctor Anthony Meyer, M.D., tests a new laparoscopic version of the UNC AR system on a patient phantom.
Image courtesy University of North Carolina*

Once these problems are reduced, the UNC team “envision[s] the introduction of a system such as [theirs] into the operating room.”

The next step will be to take advantage of the fact that the scanned data exists in the computer and use that data to automatically detect features such as lesions, cysts, etc. Pointers to the location of these features can be added to the augmented view.

In closing, the system developers report that “the system is sufficiently robust and accurate for the physician to report that the procedure on the breast model was easy” [State, Livingston, et al.].

Dr. Fuchs has even been re-encouraged in the possibility of fetus ultrasound imaging after seeing the results of another researcher comparing the face of his yet-to-be-born child with the baby’s face after birth [Sakas and Walter].

CASE STUDY 5.5: VIRTUAL REALITY EXPOSURE THERAPY

Executive Summary

The treatment of psychological disorders might not be the first thought that comes to mind when looking for ways to apply a new technology like virtual reality. However, researchers have found success in treating patients with overwhelming fears using a technique called *exposure therapy* in virtual reality environments. Early successes have spawned a number of commercial entities that supply software, and turnkey virtual reality hardware systems that can be used in therapists' offices resulting in better treatment of phobias, and lower costs.

This section looks at applications related to fear of flying, fear of heights, fear of spiders, and treatment of post traumatic stress syndrome.

Introduction

Phobias, or persistent illogical fears of some particular thing, can range from minor nuisances to devastating

disorders. Phobias can result in suffering and embarrassment to those who have them, and can have financial ramifications that reverberate throughout the economy. For example, it is estimated that 10–25% of the general population fear flying [Roberts, R.J.]. In 1982, the estimated cost of fear of flying to the U.S. travel industry was 1.6 billion dollars.

Because of the prevalence of the fear of flying, as well as the costly consequences of that fear and the high cost of traditional therapy, Professor Larry Hodges and his colleagues at the Graphics, Visualization, and Usability Lab (GVU) at Georgia Institute of Technology (Georgia Tech) began a collaboration with clinical psychologist Barbara Rothbaum from Emory University School of Medicine to develop programs that integrate virtual reality technology to aid patients in overcoming their fears. They started by addressing acrophobia (the fear of heights), and then expanded their work to include aerophobia (the fear of flying) and other disorders.

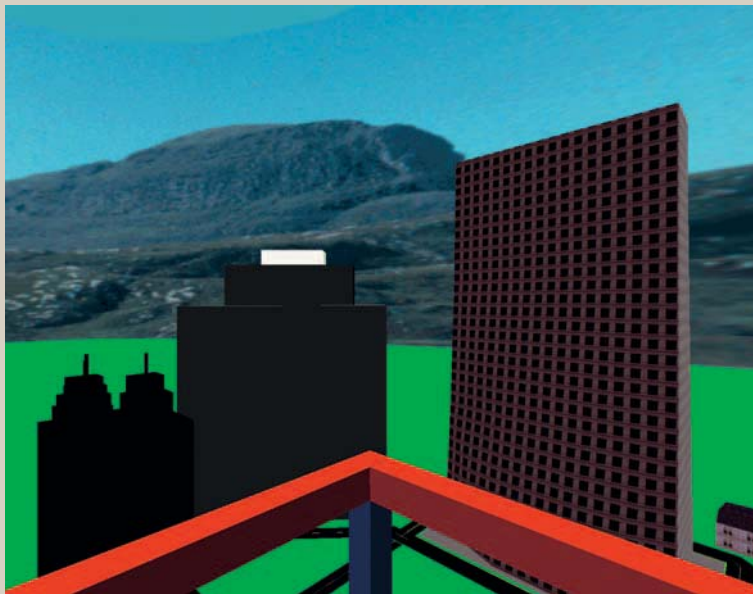


FIGURE 5-30

Patients wearing a head-mounted display are able to look off of a balcony in a virtual world to help them overcome a fear of heights. The virtual world is modeled as realistically as possible within the constraints of the rendering system.

The goal of these applications was to investigate whether VR could be effective in treating people with phobias, and to create a system suitable for use in a therapist's office. The first issue to investigate was whether or not fears would transfer into the VR environment, and whether the effects of the treatment would transfer back to the real world.

Phobia treatment is typically done by what is known as *exposure therapy*. This treatment involves exposing the patient to the anxiety invoking stimulus, and allowing the anxiety to subside. Virtual reality exposure therapy exposes the patient to a virtual environment that contains the fear-inducing stimulus rather than exposing them to the stimulus in the real world or having the patient imagine the stimulus.

A hypothesis of the Georgia Tech team predicted that if the virtual reality exposure therapy was effective, then it could be used to provide a lower-cost treatment. The major factor in reducing cost comes from allowing treatment to occur right in the therapist's office. This method also provides the therapist with greater control over the degree of stimulus exposure. If required, the VR treatment could be followed up with traditional exposure therapy by traveling to a real world location that provides the anxiety producing stimulus.

Prior to the *Virtual Reality Exposure Therapy* program, phobias were treated in two primary ways. One method is to have the patient imagine the anxiety-invoking event. The other method is "in vivo" therapy in which the patient is exposed to the actual stimulus. In both cases a technique of habituation is used where the patient is gradually moved from less threatening to more threatening situations.

Rothbaum and colleagues point out several of the drawbacks to in vivo exposure therapy [Rothbaum, Hodges, et al.]. It can be very expensive. Because standard in vivo exposure therapy involves both the patient and the therapist being in the fear-invoking situation, it requires time away from the therapist's office, and the time required for traveling to the site. These therapies can be very difficult to arrange logistically, which can also add considerably to the cost. With the fear of flying example, it would require

that the therapist and the patient travel to an airport, and that an airplane be available for their exclusive use and that they have a pilot available to fly the craft. Even if these details could all be arranged, they would be costly, and still not every scenario could be created at the therapist's command. For example, it may not be possible to spontaneously allow the patient to experience flying in a thunderstorm. Virtual reality exposure therapy addresses these limitations.

Another risk of in vivo therapy is that the patient may feel embarrassed about the treatment, and it may not be possible to maintain their confidentiality. Additionally, the phobia may be so great that the patient is unwilling to even try the treatment because they would have to face the fear directly, even though under controlled conditions. The researchers found that many people who were afraid to try in vivo therapy were willing to try VR exposure therapy. A possible explanation is that they realize that they are not actually facing the stimulus that causes their anxiety.

Application Description

The Georgia Tech research team created multiple distinct applications. The first was for the fear of heights, followed by applications for the fear of flying and other phobias. In all cases the patient wears a head-mounted display with headphones to expose them to the sights and sounds of the environment that cause them anxiety. The therapist can see and hear the world on a monitor while the patient is experiencing it. They can also see the reaction of the patient. The therapist is able to alter what is happening in the virtual world at any given moment to provide the optimal environment for aiding the patient in overcoming their fear.

The application that deals with fear of flying was more difficult to design than the fear of heights application, but the potential payoff was also greater in that it more fully exploits the concept of allowing exposure therapy that might otherwise be impractical. That is, it is usually more difficult to arrange an airplane for exposure therapy than to take someone to a high place.

Representation of the Virtual World

The items in the virtual worlds of both these applications were represented as realistically as possible. One of

**FIGURE 5-31**

The fear of flying application allows a user who is afraid of flying to wear a head-mounted display and to virtually sit in and look around a virtual airplane. The user experiences the airplane as it sits on the runway, taxis, flies in both normal conditions and storms, and lands. The virtual world is modeled at life-size.

the requirements for success with this type of therapy is that the participant be mentally immersed in the environment. That is, they need to believe the situation in order to feel the anxiety as they would in the real world. The developers felt that greater realism would enhance mental immersion.

The virtual world in the *fear of heights* application consists of a glass elevator, a series of bridges, and a series of balconies. The elevator has virtual buttons that can be pressed to go up and down.

The virtual world of the *fear of flying* application consists of a model of the passenger cabin of a commercial aircraft and a view of the outside as seen through the window of the plane. The passenger is seated in a window seat in the cabin, and can see appropriate views out the window both on the ground and in the air. Thus, the various airport buildings, and the runways and taxiways were modeled. While flying, the passenger can see the ground, and/or clouds.

The worlds are represented internally as polygonal objects. Specifically, they were stored as Wavefront™ object files.

Both the *fear of heights* and *fear of flying* environments were created to look as realistic as possible and to emulate the kinds of details one would find in the environment. Additionally, both environments were created to life-like scale as observed from the first-person point of view. In the *fear of heights* application, the elevator was a glass elevator that overlooked the virtual interior lobby of a virtual hotel (inspired by the inside glass elevator in the Marriott Marquis in downtown Atlanta). The participant's hand and arm are represented in the scene as an avatar of the user.

In the *fear of flying* project, the airplane seats were texture mapped to look like the actual color and fabric of actual airplane seats. The cabin was built to appropriate scale such that the seats were the right size, the aisle width was appropriate, etc. Other details such as seatbelt signs mimicked the real-life equivalent in look, size, and placement. The windows were the correct size and in the correct location.



FIGURE 5-32

A glass elevator in a hotel lobby is modeled after the Marriott Marquis in downtown Atlanta, GA. The patient can press virtual buttons to cause the elevator to ascend and descend.

The participant could also see a representation of a generic body where they sat (i.e., their avatar). They see a pair of blue jeans-covered legs and a torso if they look down. The body is static, however, and is the same for everyone.

The researchers expressed a belief that sound in the virtual environment is very important, and is a big part of the virtual therapy experience.

The early work done on the *fear of heights* application, however, had no audio representation. The *fear of flying* application made use of a variety of sound effects. The sound effects can be categorized into two basic types. One type of sound is continuously looped. Sounds like the engine fall into this category. The other type is triggered as a one-shot sound effect. Examples of this are the prelanding and tire sound effects. The sounds were coordinated with the visual display to enhance the sense of realism.

The sounds were played on a machine other than the visual rendering machine, to prevent the computational load of visual and sound rendering from competing for the

same resources. The sound computational system could play multiple sounds simultaneously, allowing for a richer sound experience. More recently, PCs are now capable of handling the rendering of both sight and sound from a single (relatively inexpensive) computer.

In the two cases discussed so far, the only type of haptic feedback provided was via static physical objects. The fear of heights experience had a waist-high rail that a participant could hold or lean against, providing haptic feedback as well as ensuring the participant stayed within range of the tracking system. This aided greatly in the application, as it allowed a participant to lean over it and look down. Additionally, the platform on which the user stood was raised a few inches above the floor. Thus if users stuck their feet over the ledge, they could feel edge, enhancing the realism. In the prototype version of the *fear of flying* application, the participant sat in a normal chair with no seatbelt. However, in a later version that was commercialized, a ThunderChair™ seat with an added seatbelt was used.

For some experimental phobia treatment applications, the researchers experimented with the use of a subwoofer to provide vibration sensation to the participant. This was originally done by mounting the speaker to the bottom of the seat on which the participant sat. Later, a commercially available product, the ThunderChair™, was incorporated. The ThunderChair also uses the technique of using subwoofers to produce vibrations throughout the seat and back of the chair.

There was no vestibular feedback in either of these applications. The decision to not use vestibular feedback was made to help keep the system practical for deployment in therapists' offices, both from a cost and size point of view. The research team does not feel the lack of vestibular feedback is significant, and that the effects they are after can be achieved through the visual display.

Neither application integrated odors in the environment. The developers of these applications believed that, like sound, smell is potentially an important component in VR systems for treating phobias. Thus, in a separate project they studied the effects of other sensory stimuli such as smell, wind, heat lamps, etc. In their study, Dinh and colleagues put 322 subjects through a virtual environment consisting of a small office suite [Dinh et al.]. Subjects were exposed to different sensory conditions—the inclusion of smell or not, the addition of sound cues or not, the addition of tactile cues or not, and high versus low visual resolution. The results indicate that the addition of more senses increases the reported feeling of presence and ability to recall aspects of the virtual world. This benefit is particularly strong for the auditory and tactile cues.

Interaction with the Virtual World

These applications fall into the medical (psychiatric) therapy genre. The form is typically to mimic real-world interactions as much as possible, allowing the patient to interact and react as intuitively as in the real world.

The narrative in these applications is that the patient experiences a real life situation, in which some events are controlled by a therapist. In the *fear of heights* application, patients get on an open elevator and ride it up and look around. They then progress to other experiences such as standing on a balcony. Of course, the therapist eases the



FIGURE 5-33

A patient experiencing the fear of heights virtual reality application presses an elevator button. Note the tracking sensor on the participant's hand, and the railing he leans against, emulating the railing in the virtual world.

patients into the various steps in the procedure, which may take many sessions.

In the *fear of flying* application, the full sequence of events consists of the patient sitting in the airplane parked at the gate, taxiing to wait at a holding position on the runway, taking off, steady flight, flight with various special occurrences such as darkness or storms, and finally landing. In both applications, time passes at the same rate as it does in the real world.

Travel through the virtual worlds is based on the specific phobia being treated. In the *fear of heights* application, the interface is physical locomotion. The participant can also press the buttons on the elevator to select how high they go. In the *fear of flying* application, the ride-along method of travel is used, mimicking how most of us travel by air. The airplane follows a prescribed spline path with the patient remaining seated the entire time.

In both cases, the therapist is in control of moving the scenario from one stage to the next. In early aerophobia (fear of flying) sessions, the patient may spend the whole session sitting in the stationary plane or taxiing around the runway. In the acrophobia (fear of heights) treatment, the therapist transitions the patient from scene to scene (e.g., the elevator to the balcony) as appropriate.

The therapist has a user interface that allows them to control the simulation in a very simple way. Because the therapist's first priority is to pay attention to the patient, a complicated (e.g., menu-driven) user interface would be detrimental. Thus, the interface consists only of a three-key interface on the keyboard. The actions of those keys are context sensitive, based on the current state of the system. A state machine dictates which situations are possible from any given point in the simulation.

When the patient is ready, the therapist can quickly induce a change in the state of the system (e.g., transition from taxiing to taking off). Each state is maintained long enough to be convincing, but not so long as to disrupt the normal flow of the therapy. A state will remain in effect until the therapist takes action to change it. For example, the airplane in smooth flight will remain in smooth flight until the therapist chooses to have something new happen. The therapist may end the simulation at any time.

There are no objects to manipulate in either simulation except for the elevator buttons in the fear of heights system. In that case, the hand is tracked by a tracking sensor to sense contact between the user and the virtual buttons.

There is no agent interaction in either simulation. The virtual worlds are not populated with other people. There is a sound effect of a virtual pilot making announcements such as when it is time to land, but there is no interaction with the pilot.

All motions in these simulations were preprogrammed, choreographed motions. The therapist changes scenarios by invoking a different state in the system. For example, in the *fear of flying* simulation, the aircraft proceeds from taxiing to takeoff only when the therapist signals this to occur. In flight, the therapist can activate or deactivate turbulence while monitoring the subject's level of discomfort. In yet another application, a virtual Vietnam War experience,

presentation of gunfire or radio chatter occurs only when the therapist activates the appropriate buttons.

Venue

The venue for which these applications are designed is the therapist's office. With this in mind, when doing a case study analysis of the project, a space similar to a therapy office was used at GVU. The facility has a specially constructed room with a one-way glass through which researchers could watch subjects. The site was equipped with video cameras so they could capture the sessions for future study. The therapy room was accessible via its own entrance so subjects would not need to walk through the main part of the laboratory.

Throughput of the *Virtual Reality Exposure Therapy* system mimics that of a therapist—one patient has access at any given time, with sessions lasting approximately 1 hour.

VR System

The prototype system was centered around a Silicon Graphics Reality Engine™ workstation as the main computer. This system provided the visual rendering used by the applications. The project has used several different head-mounted displays over the course of time. The *fear of heights* application used a Virtual Research Flight Helmet™. The original *fear of flying* application used the Virtual Research VR-4™.

The software was ported to a PC-based system for lower cost under a grant from the state of Georgia. This version used the Virtuality Visette™, which offers one-quarter NTSC resolution. The developers also tested some participants in a monocular Liquid Image MRG-4™ HMD. Later, the developers began to use and recommend the VR-6™ HMD from Virtual Research, Inc.

The images are displayed monoscopically. The decision to not use stereo image display was based primarily on the desire to expend their computational resources making a single higher-quality image than to create two lesser-quality images. The developers calculated the possible enhancement of the stereoscopic depth cue based on the resolution of the HMD and determined that it would not add much information for either the *fear of heights* or *fear of flying* experiences. Therefore they chose to create a more complex world instead.

Both applications use 6-DOF head tracking. In an effort to keep costs as low as possible, they opted to track only the head for most phobia-treatment applications. One exception is the *fear of heights* application, which uses an extra tracking sensor so the system could tell when the hand pushed the elevator buttons. Position tracking can be done with either the Polhemus or Ascension electromagnetic systems.

In some cases, monitors of the patient's heart rate and other body functions are used to allow the therapist to have an objective measurement of the patient's physiological responses.

Application Implementation

The *Virtual Reality Exposure Therapy* project was conceived late in 1992. The first code for the *fear of heights* application was written in the spring of 1993. By October 1993 they ran their first patient in VR therapy. For the *fear of flying* application they began taking measurements of the aircraft cabin in the summer of 1994. The cabin and outdoor model was built through the rest of 1994. The simulation program was written from January to summer of 1995. In the summer of 1995 they ran a complete case study with an aerophobic patient.

The system was developed in-house by a team that included psychiatrists and computer scientists. The study was conducted in facilities provided by the GVV lab at Georgia Tech. The project was led by computer science professor Larry Hodges at the Georgia Tech GVV, and psychologist Barbara Rothbaum of the Emory University School of Medicine. Other investigators who contributed to the design of the study were Dr. Major James Williford (U.S. Army psychiatrist who was stationed in Augusta, Georgia, at the time), Dan Opdyke (Georgia State University), and Max North (Clark Atlanta University). Rob Kooper programmed the initial application, with other programming contributions later made by Kevin Hamilton, Ben Watson, Jarrell Pair, Brian Wills, David Gotz, and Phillip Hebert.

The initial software was written using the Georgia Tech *Simple Virtual Environment* (SVE) Toolkit which is based on *OpenGL*. Because of the lack of availability at the time for good *OpenGL*-based graphics accelerator cards for PCs, the initial port of the *fear of flying* application used the *DirectX/3D* library. Later PC versions used *OpenGL* on a *Windows NT* platform. By using *OpenGL*, the developers

could more easily transfer applications from the faster Silicon Graphics workstations to the cheaper personal computer platforms.

To create the model of the cabin, a team of two undergraduate students and a high school teacher working as a summer intern at the GVV lab were allowed access to a Boeing 747 to take photographs and measurements. This process took several hours to complete. These measurements were then used to create the model using Wavefront Inc. modeling software.

The sounds used in the *fear of flying* application were obtained from commercial sound effect CDs and converted to computer sound files. To be useful in this application, these sounds had to be edited for length and content.

Related Applications and Follow-on Work

Other research groups have also been investigating the use of VR for exposure therapy. About the same time the Georgia Tech project was being prototyped, Dr. Ralph Lamson at the Kaiser Permanente health clinic in San Rafael, California, also had the idea to apply VR for helping sufferers of acrophobia. He found that virtual reality was effective in relieving the fear of heights in an experimental group of subjects [Lamson].

At the University of Washington in Seattle's Human Interface Technology Lab (HIT Lab), Hunter Hoffman has done some interesting work on arachnophobia (fear of spiders). In his work, Hoffman used tracked, physically manifested objects to transfer object permanence from these selectively enhanced objects to the rest of the objects in the virtual world, making that world more mentally immersive. In this case, Hoffman attached a prop composed of a fuzzy ball with multiple pipe-cleaners to mimic the body of a large spider. The prop is tracked, and thus is located in the same place in both the real and virtual worlds. The user can therefore feel the spider simply by reaching out to where it appears to be in the virtual world. [Hoffman, VRAIS '98]

Hoffman reported great success when his application was tested on the first patient. This patient, a Seattle-area woman, had a long-term fear of spiders that had grown so extreme that she vacuumed out her car every day before using it and placed a towel in her bedroom door jamb every night to avoid spiders. After the therapy, the woman's fear

**FIGURE 5-34**

Researchers at the Human Interface Technologies Laboratory at the University of Washington created a virtual reality application to treat arachnophobia (fear of spiders). In this application, they provided a physical model of a spider consisting of a fuzzy body with pipe cleaner legs that is colocated (via a tracking sensor) with the virtual representation of the spider. Thus, a patient wearing a head-mounted display can reach out and touch the virtual spider and feel the fuzzy physical object.

Top: Courtesy of Hunter Hoffman. Bottom: Courtesy of the BBC

had been reduced to the point where she was able to go camping.

Conclusion

The subject for the *fear of flying* case study experiment was a 42-year-old female who had a debilitating fear of flying [Rothbaum, Hodges, et al.]. Her fear had increased over

**FIGURE 5-35**

Virtual Iraq is a timely application of virtual reality. Dr. Albert "Skip" Rizzo modified a video game to create a virtual reality application that is used for treating victims of post-traumatic stress disorder. A group of collaborating researchers, including Virtually Better Inc., developed it and are involved in evaluating its effectiveness.

about 5 years before the therapy took place, culminating in her avoiding flying completely for the previous 2 years. This fear resulted in her missing out on things in her life such as vacations and business opportunities that required travel. Since she was embarking in a new business venture requiring travel, she sought the VR therapy hoping that it would help her get over her fear of flying—mainly the fear of crashing.

The subject completed a battery of pretherapy questionnaires, followed by seven sessions (as a paying patient) of outpatient therapy in which she was taught anxiety management techniques. Because it appeared that it would be beneficial to all parties involved for her to undergo exposure therapy, she was invited to participate (at no cost) in the *VR Exposure Therapy* program.

Once again, the patient was given the same set of questionnaires. The patient was then allowed to progress at her own pace in a series of sessions which lasted about

35–45 minutes each. Through the course of the six sessions, the patient encountered various scenarios such as taxiing on the runway, smooth flight, turbulent flight, etc. The therapist was able to observe the patient and see the view she saw. Thus, the therapist was able to aid her in overcoming her fear through appropriate encouragement and other techniques. During the testing the patient gave subjective ratings of anxiety, ranging from 0 (calm) to 100 (panic).

In addition to using standard testing questionnaires, the research team developed two forms of their own. One was a “flight self-monitoring sheet” on which the subject provided ratings for different parts of the flight experience. The other was a form that the therapist used to record what VR situations were presented to the patient. The patient filled out the self monitoring sheet several times during the testing, after the therapy, during an actual flight she took after the therapy, and again a month after the therapy had been completed. At that point she also took the original battery of questionnaires again.

Two days after the therapy, the patient successfully completed a cross-country flight with her family that she was scheduled to make. She reported that it was comforting at points to remember and picture the VR experience. Although the therapy was considered successful, the exact amount of benefit from the VR cannot be precisely determined because she had also been treated with anxiety management therapy.

In addition to the above case study, virtual reality exposure therapy has proven effective enough to move from the research lab to the marketplace. Because the *fear of flying* application generated considerable success stories and press interest, the research group received more queries about using their system than they could address. In response to this, Hodges and Rothbaum have formed a company known as Virtually Better, Inc. (VBI) to distribute the software to interested parties. However, instead of selling the hardware directly, they provide a shopping list of the necessary hardware components.

VBI reported six other sites using their exposure therapy software in addition to their own clinic at the time of this writing. This did not include the other centers that independently developed their own software for therapy using VR. This large number of actual users of virtual reality exposure

therapy is perhaps the best statement of how effective VR is in this domain. Brenda Widerhold, the director of Center for Advanced Multimedia Psychotherapy at the California School of Professional Psychology (CSPP) is one user of the VBI system. She reported to treating 30 aerophobic patients in her first year of using the system, and 28 of these patients were able to fly after about eight training sessions using the VR experience. Widerhold used heart rate, respiration rate, and sweat sensors in conjunction with the VBI system to allow her to better monitor the level of her patient’s anxiety.

Mental immersion is of prime importance for the use of virtual reality exposure therapy. People treated for various phobias using VR must be sufficiently immersed in the experience to exhibit the same symptoms in the VR experience as in the real world experience. Perhaps surprisingly, many people do indeed experience the same symptoms in VR that they do in the real world.

For example, the subject of the case study described above did behave as though she was on a real flight while she was in the VR system. She experienced the shortness of breath, butterflies in the stomach, etc. that are classic symptoms of anxiety. These indicators provide evidence that the patient was mentally immersed.

Perhaps the most significant discovery in this application was that phobias do transfer into the virtual environment. Participants exhibit anxiety symptoms when exposed to their fear-inducing stimulus in the virtual environment. Additionally, the benefits of exposure in the VR environment benefit the participant in the real world situation. Interestingly, some people are willing to undergo therapy in the virtual environment who are not willing to try in vivo exposure therapy.

In addition to the two phobia treatments described here, Hodges/Rothbaum and other groups have also developed many other virtual reality environments for phobia treatment. VBI offerings included software for treating the fear of flying, heights, malls, storms, and driving. In addition to this, they have developed software to help veterans overcome post-traumatic stress disorder (PTSD). This “virtual Vietnam” application has been developed for deployment at VA Hospitals, where it can be used by therapists to allow sufferers to relive and recall events to hopefully help them overcome the disorder.



FIGURE 5-36

Virtual reality is used by therapists to allow Vietnam War veterans to experience, in a controlled environment, many of the sights and sounds they experienced in the Vietnam War. Virtual Vietnam includes a chair equipped with a subwoofer to provide the sensation of the vibrations of riding in a helicopter.

Virtual Vietnam places the veteran in a few scenarios common to Vietnam veterans. These include being transported via helicopter between their base and the battle. Additionally, they are also able to experience standing in a field as the helicopters take off and land. To give the subject a greater sense of immersion, they are seated in a ThunderChair device that helps them feel what it was like to be in a vibrating helicopter and feel the roar of the gunfire outgoing from the nearby gunner and incoming from unseen enemies.

A heart rate monitor is attached to the subjects as they experience the representation of the Vietnam War.

When used as part of an overall therapy, the virtual Vietnam experience has been a useful tool in helping veterans recall

past circumstances and overcome their associated problems [Hodges, Rothbaum, et al.].

Virtual reality is proving to be a beneficial tool for psychologists treating phobia disorders, but there is still room for improvement. As Barbara Rothbaum points out:

VR exposure therapy has many advantages for several anxiety disorders, primarily centered around the therapist control over stimuli, the ease and efficiency of therapy in the office, and the decreased expense of time and money. The major drawbacks at this point are the expense of the hardware and the expertise required to program the software, but these are both destined to change for the better in the near future.